

# Generalized Branching in Circle Packing

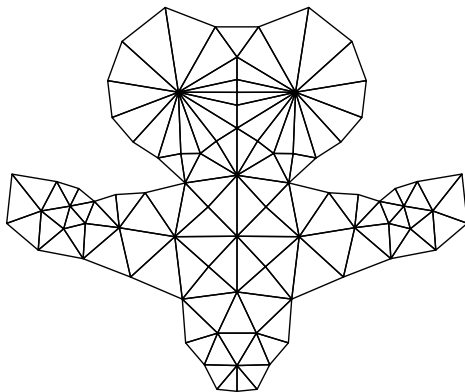
## A Discrete Ahlfors Function

Edward Crane, Ken Stephenson, and James Ashe

Johnson C. Smith University

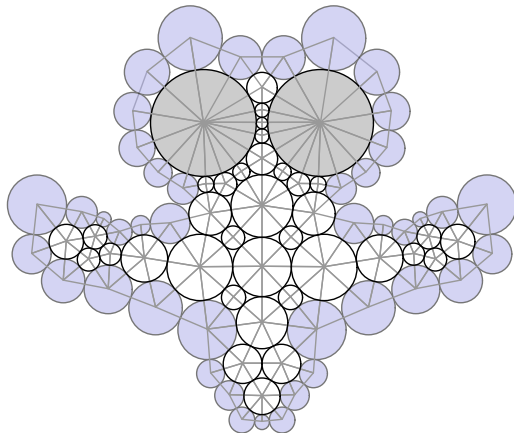


# What is a Circle Packing?



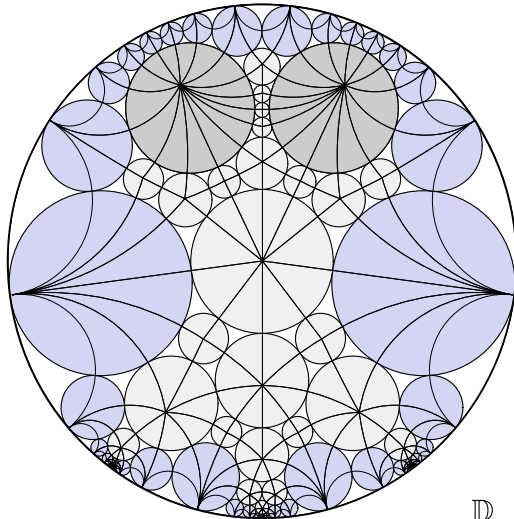
# What is a Circle Packing?

**Geometry + Complex = Circle Packing**



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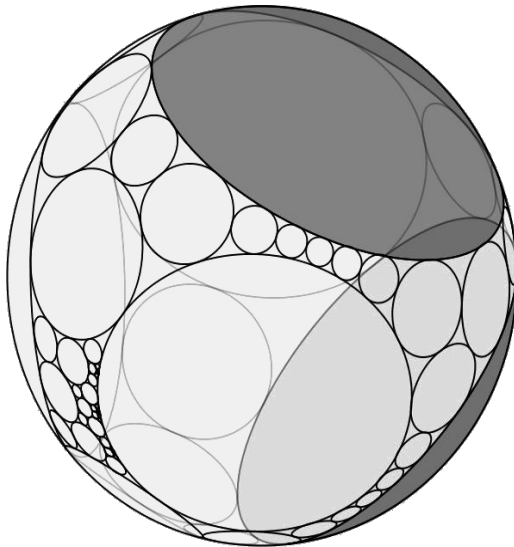


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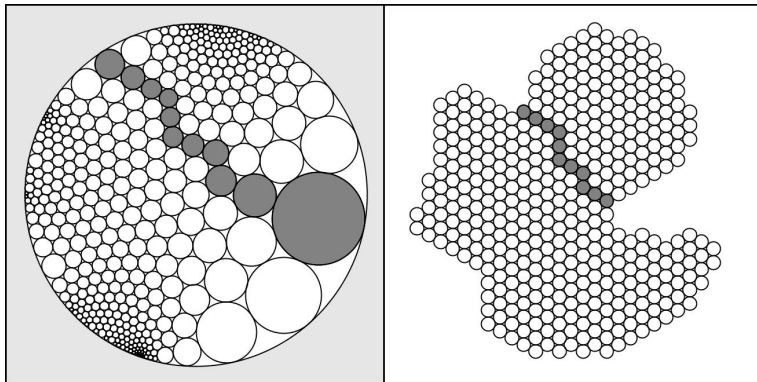


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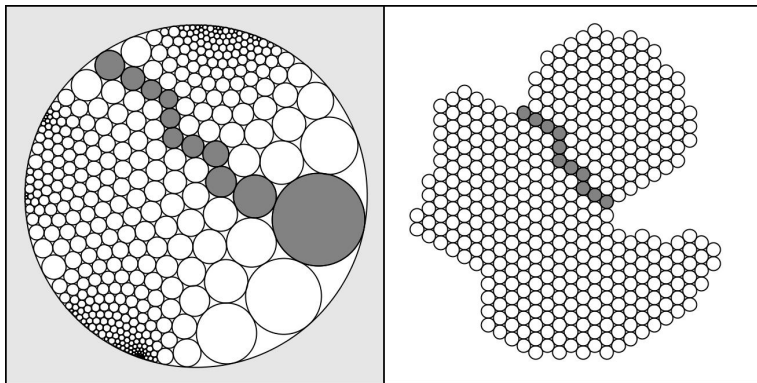


# Conformal Mapping



When simply connected.. everything works beautifully.

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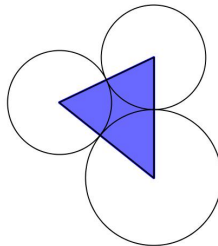


When simply connected.. everything works beautifully.  
When not simply connected the current theory does not suffice

Flowers with angle sums of

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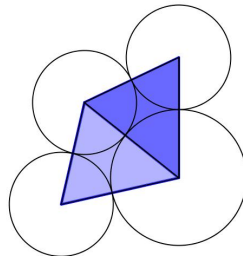
- $2\pi$ , a (trivially) branched flower.



# Local coherence and branching

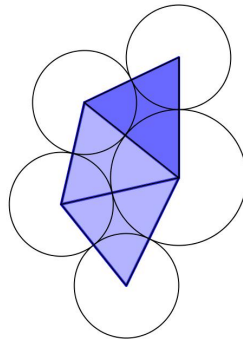
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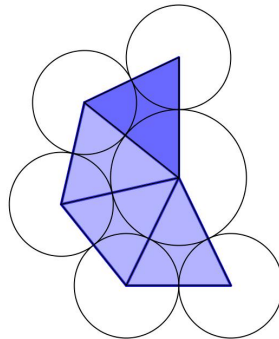
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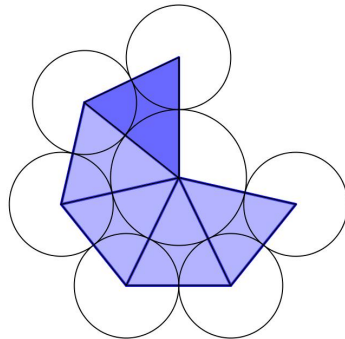




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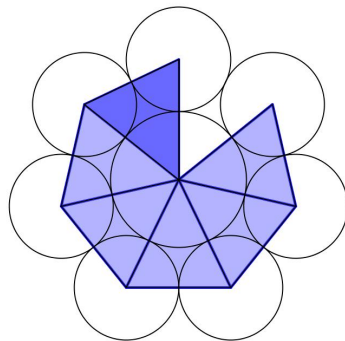
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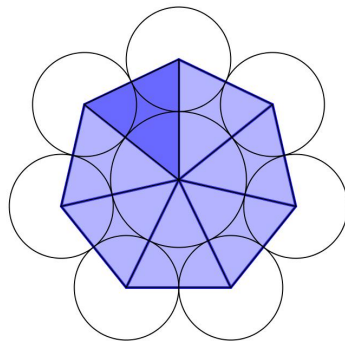
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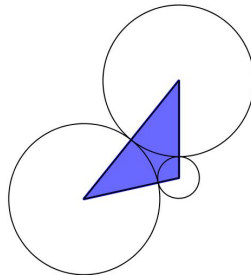
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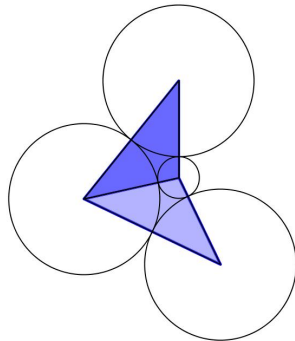
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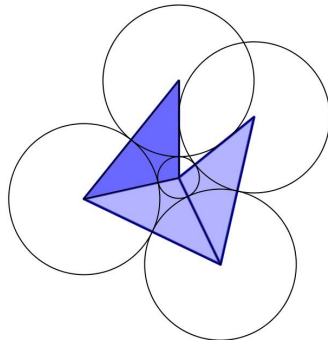
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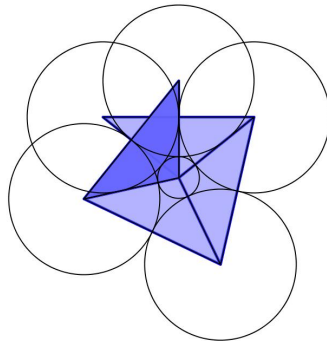
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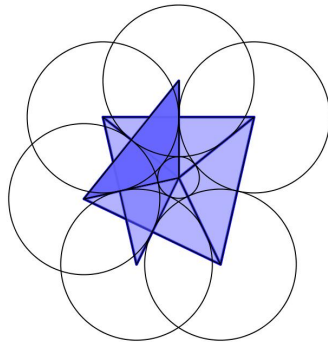
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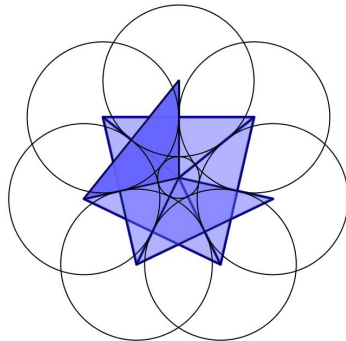




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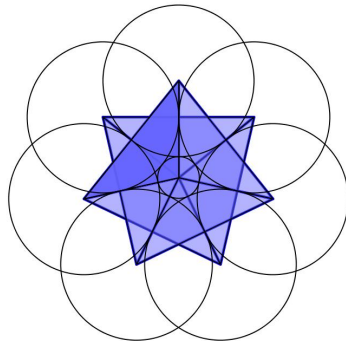
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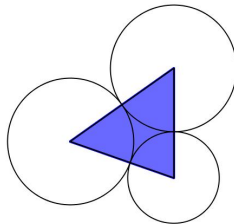
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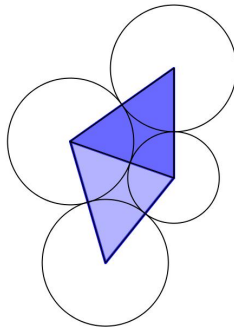
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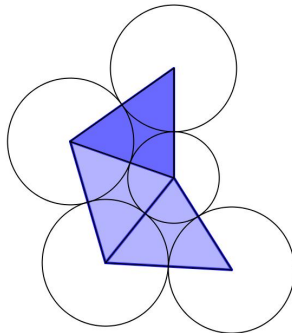
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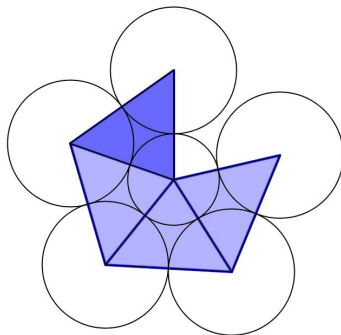
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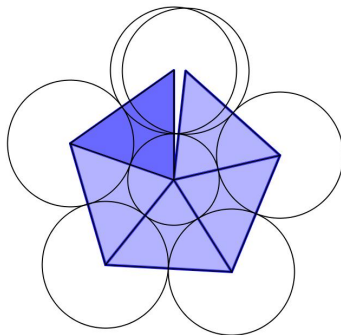
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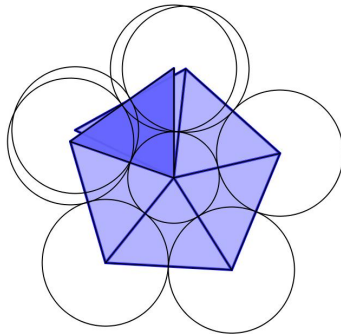
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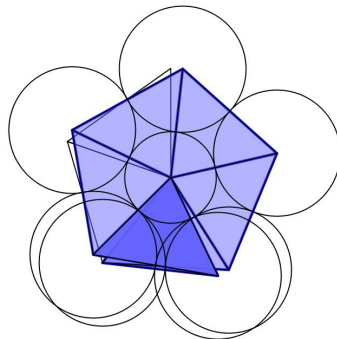
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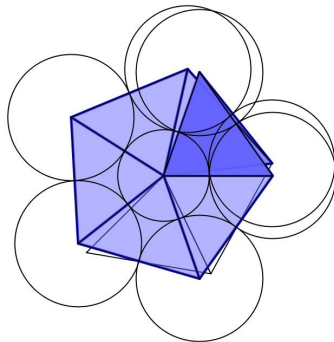
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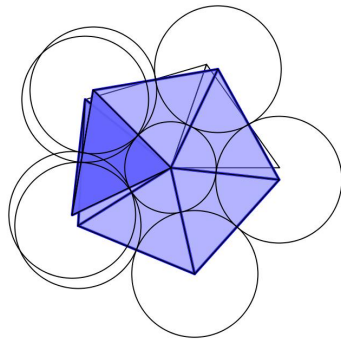
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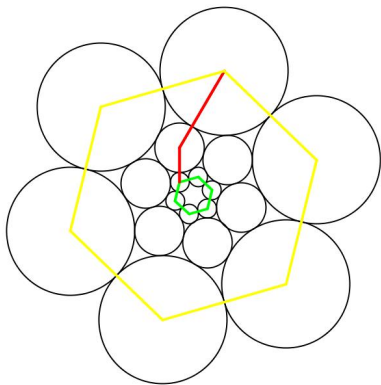


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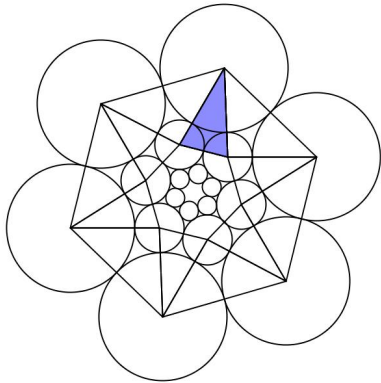
# Local coherence $\nRightarrow$ coherence



Example

A circle packing.

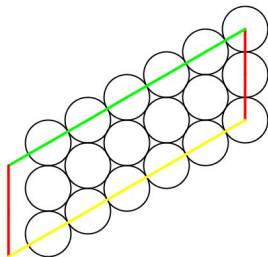
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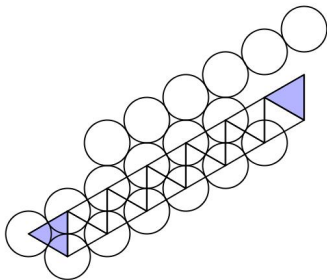
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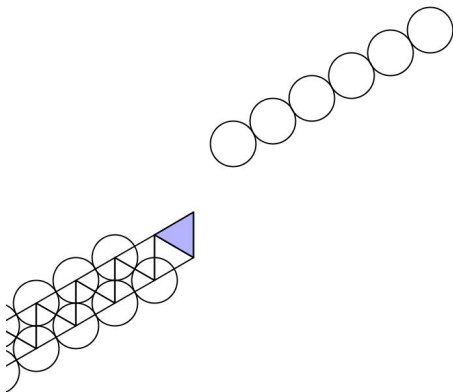
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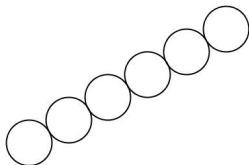


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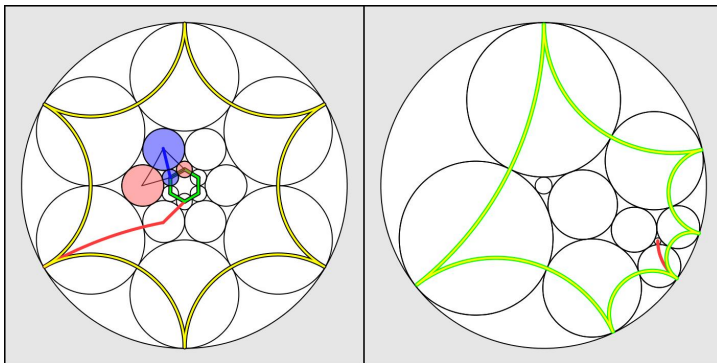


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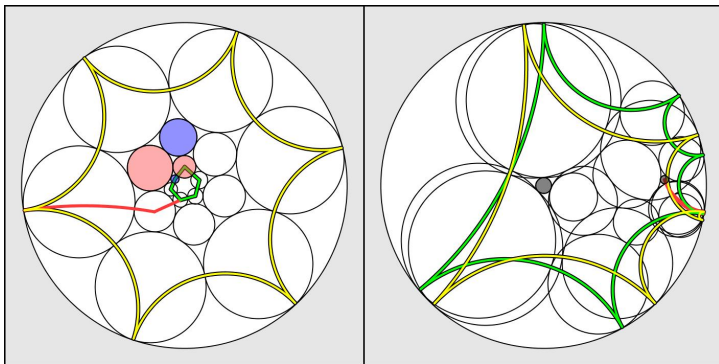
**Ahlfors function:** maximal packing of an annulus.

- What we want.



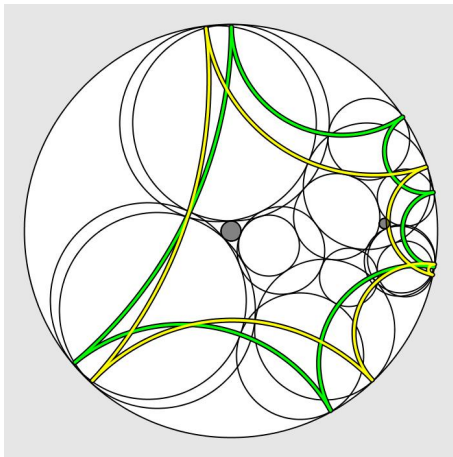
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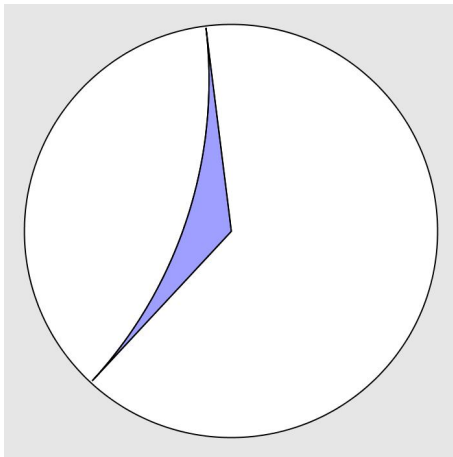
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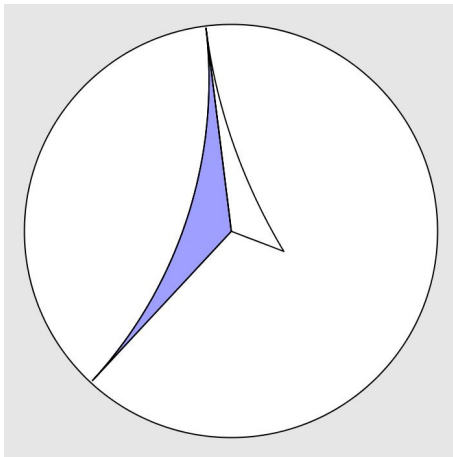
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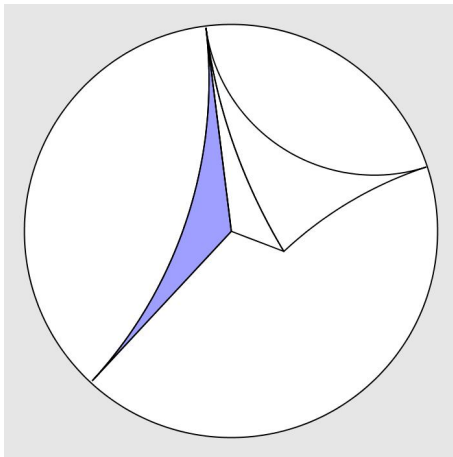
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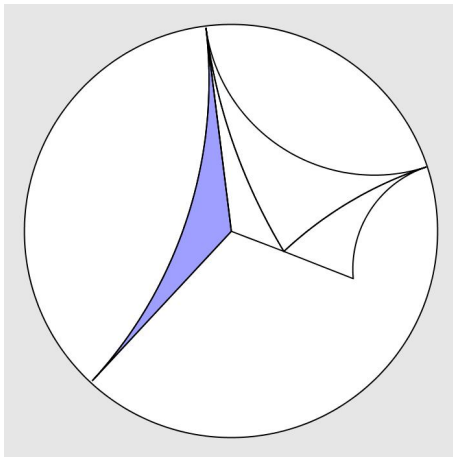
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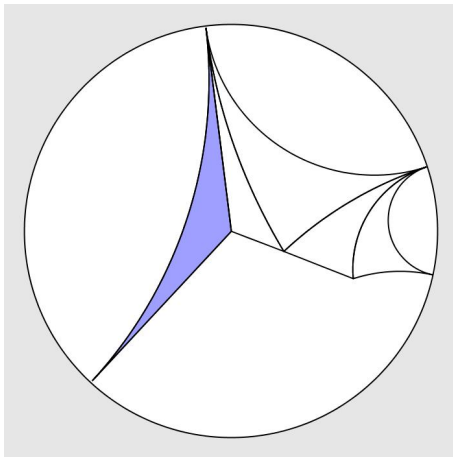
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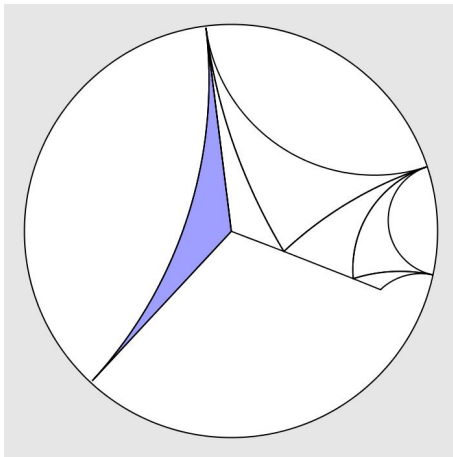
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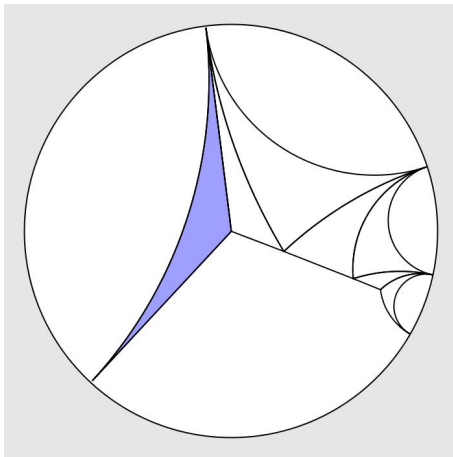
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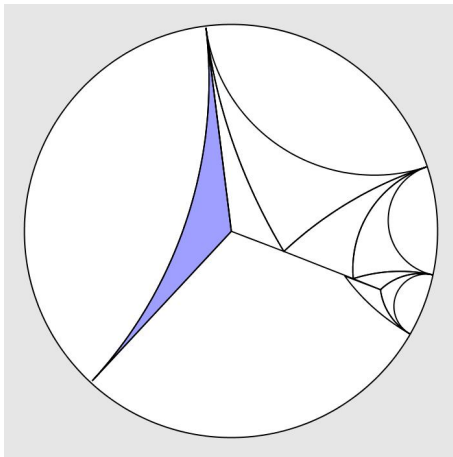
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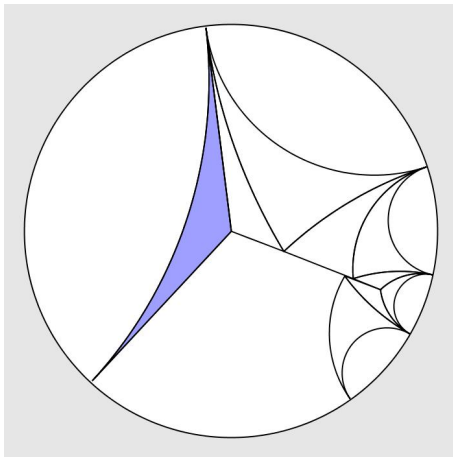
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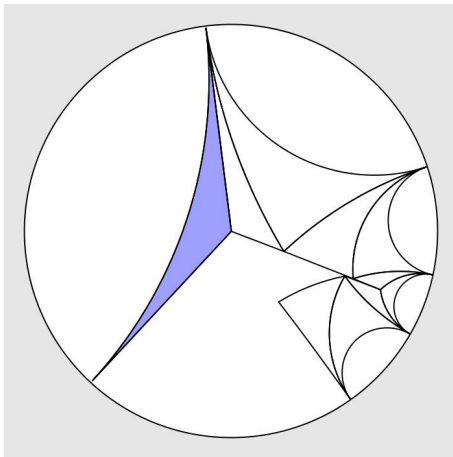
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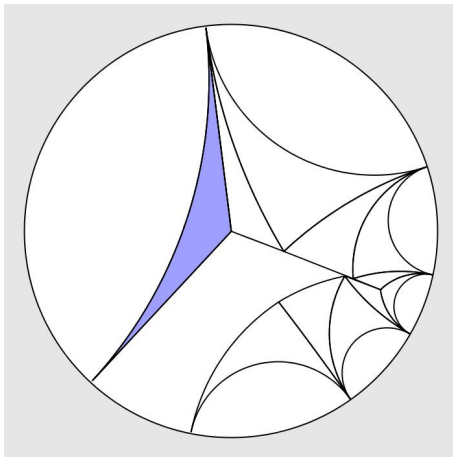
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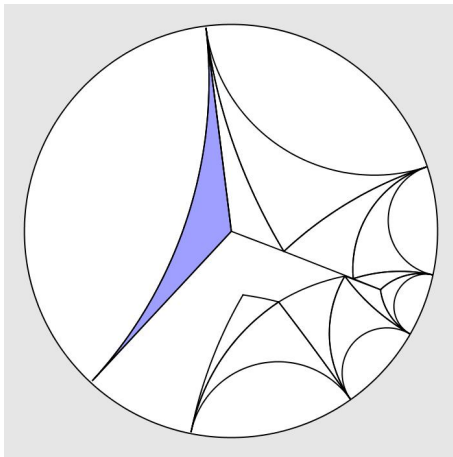
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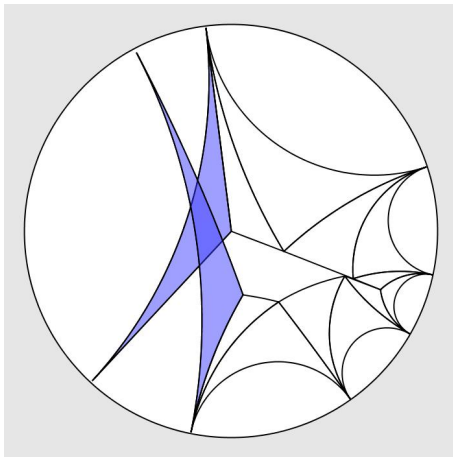
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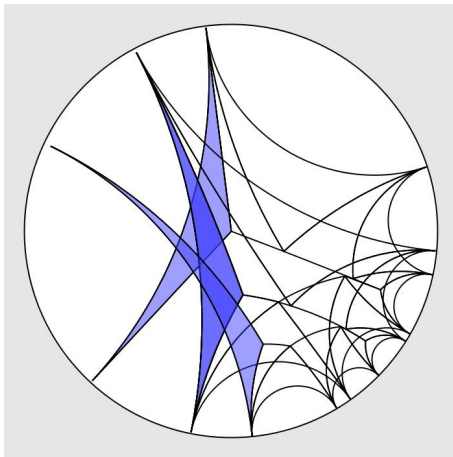
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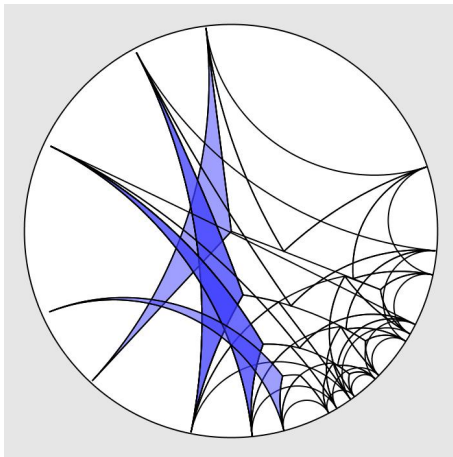
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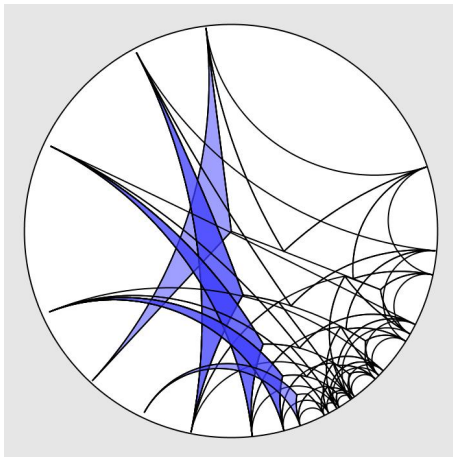
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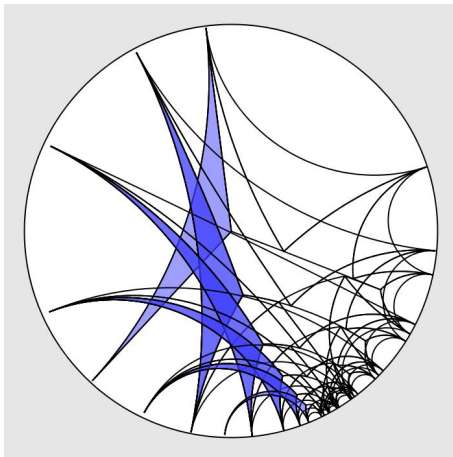
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# Generalized branching

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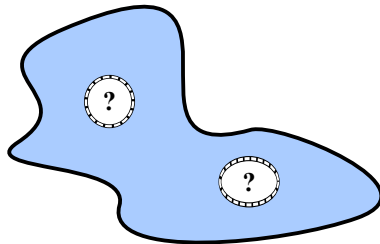
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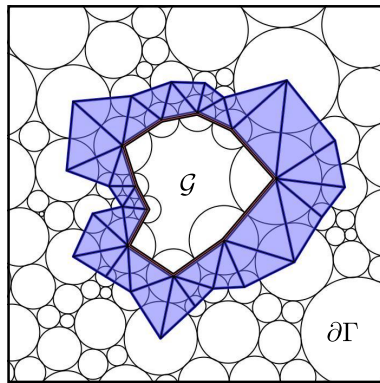


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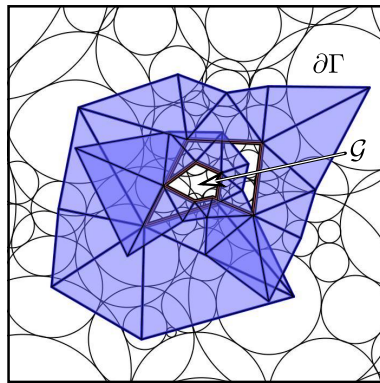


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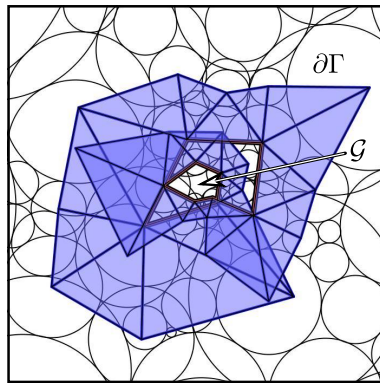


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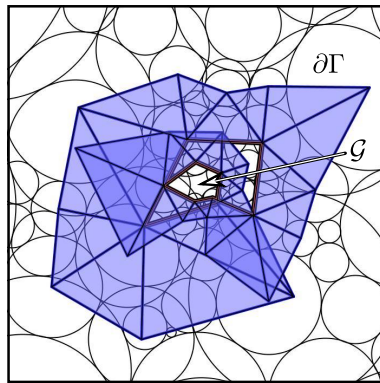


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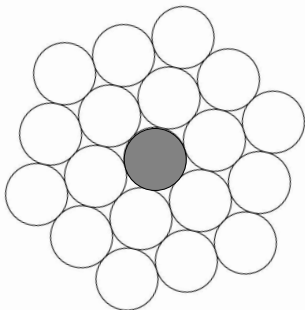
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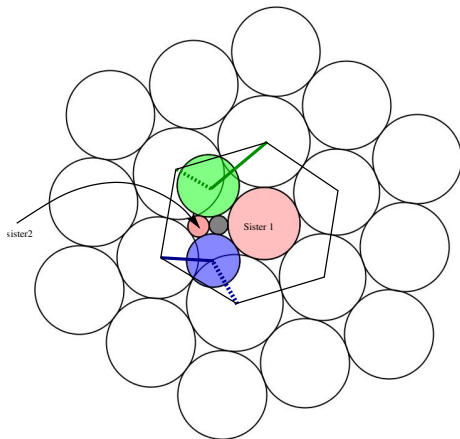


# Chaperone Branching (in a circle)



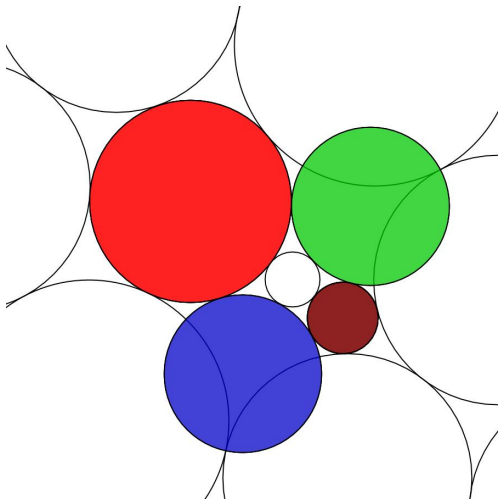
- Auxiliary circles are introduced, one at the “branch point”.
- Overlap of  $2\pi$  is distributed on each Chaperone circle.

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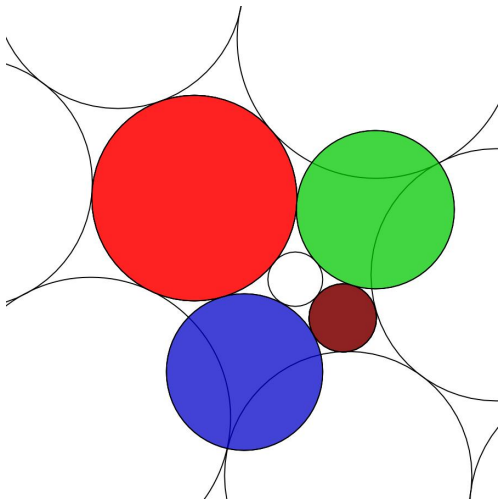
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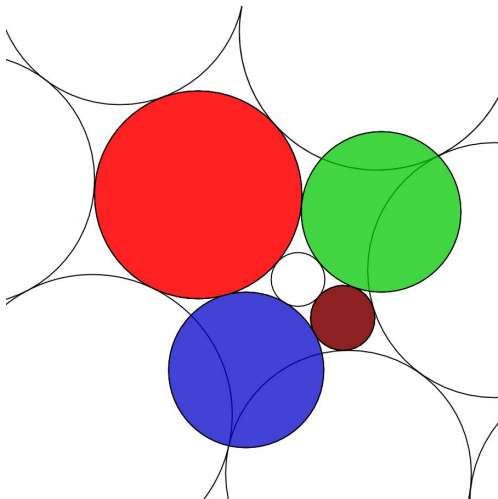
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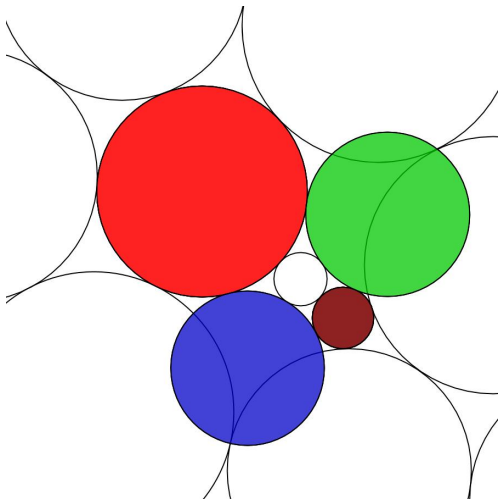
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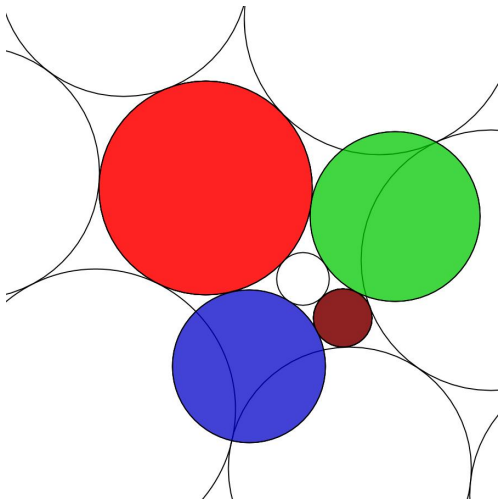


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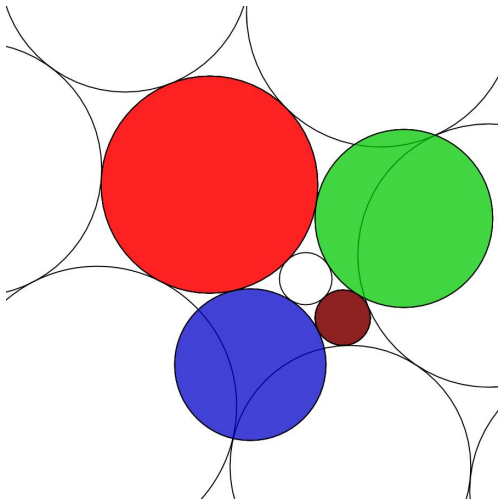
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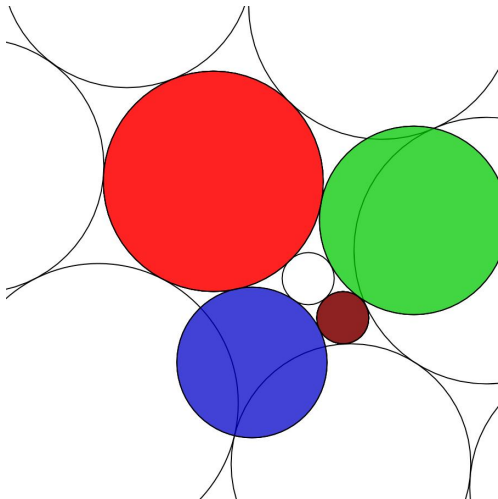
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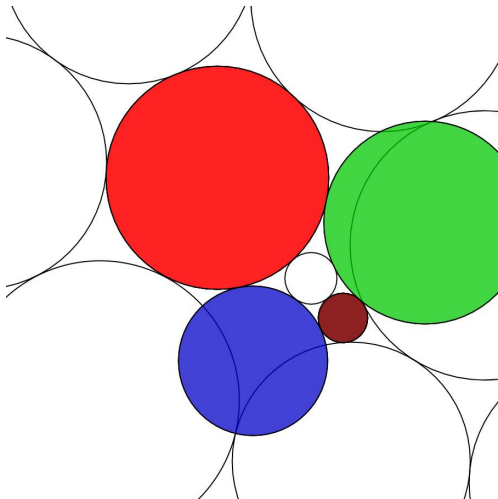
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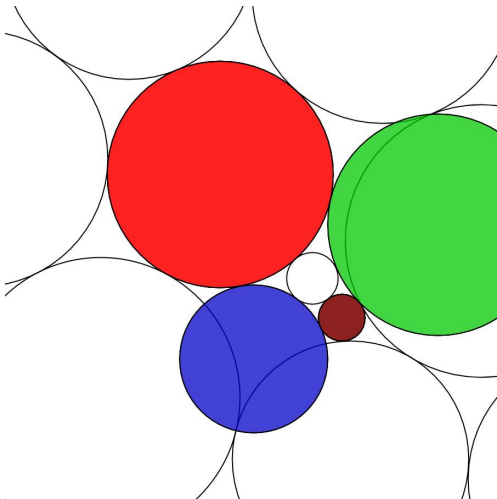
- Auxiliary circles are introduced, one at the “branch point”.
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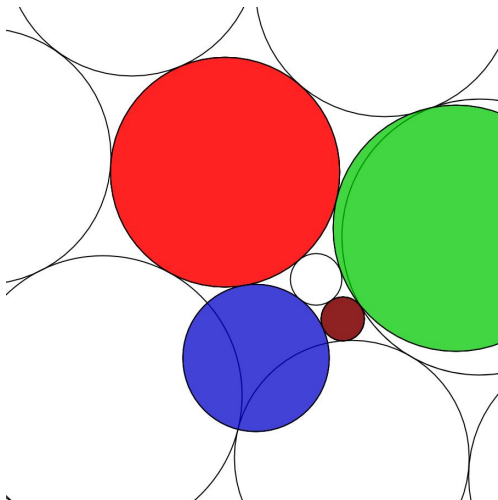
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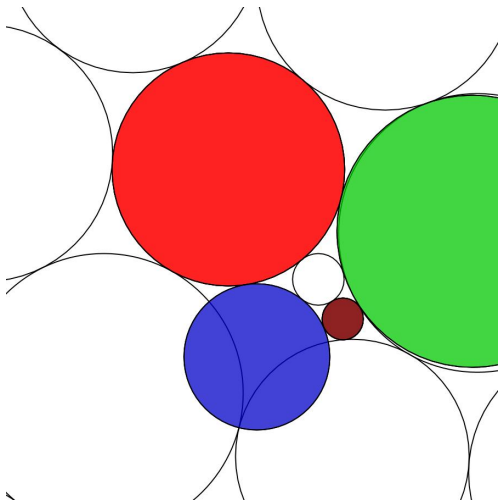
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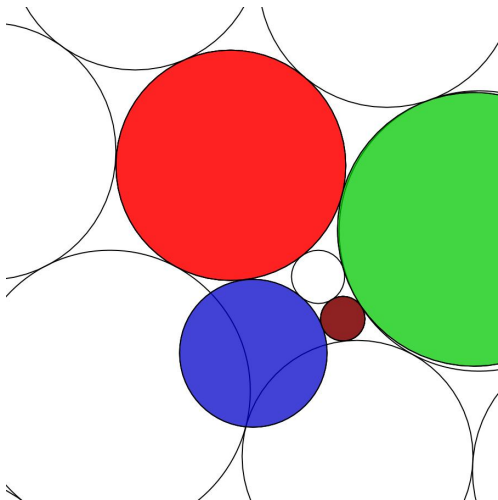
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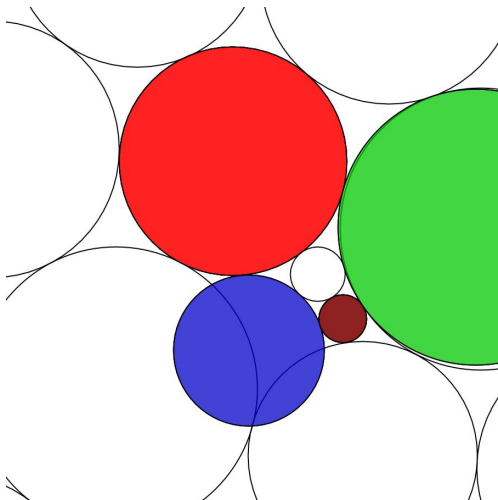


# Chaperone Branching (in a circle)



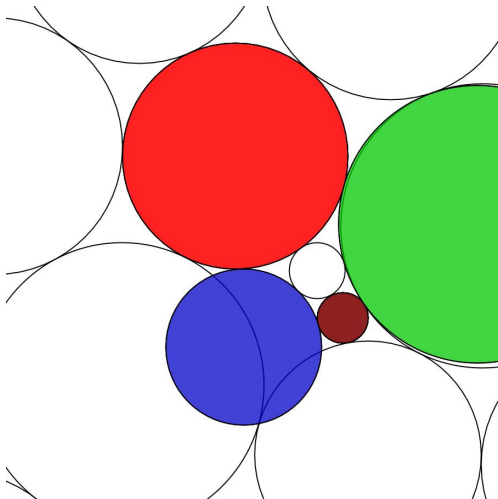
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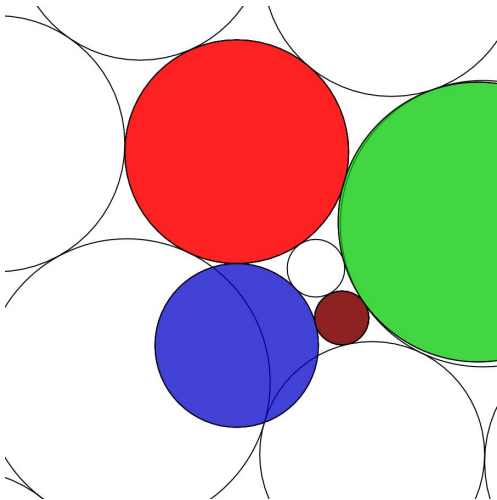
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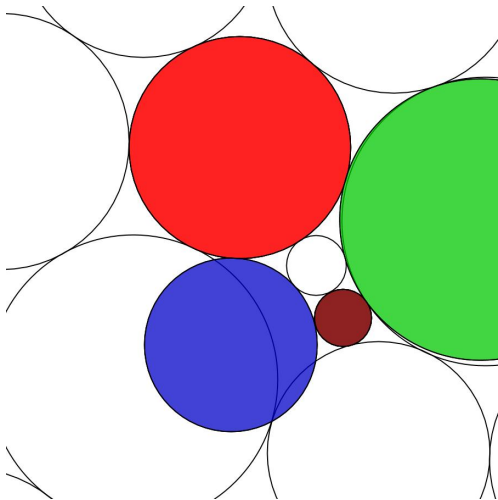
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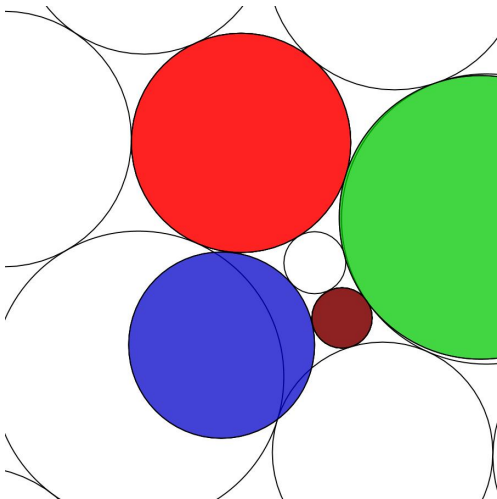
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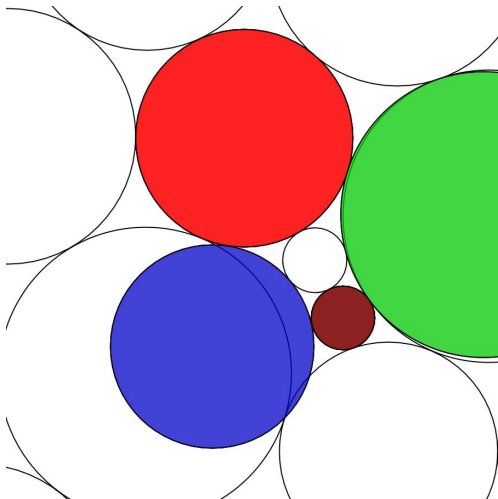
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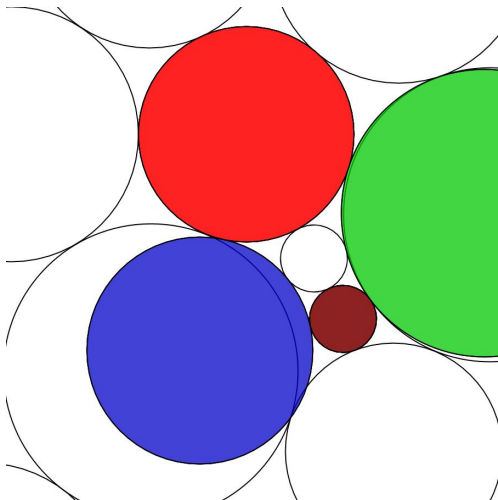
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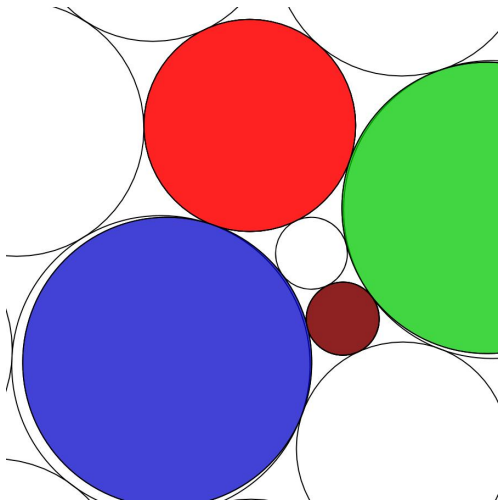
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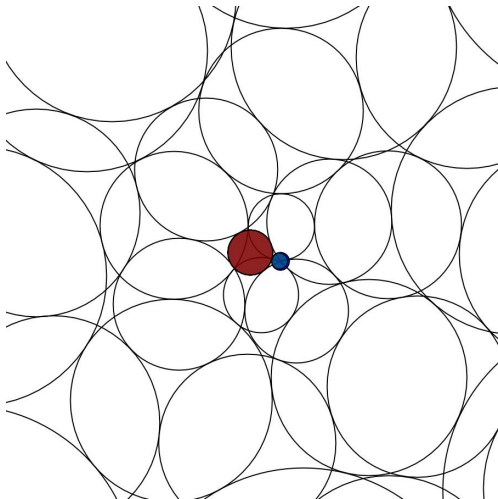


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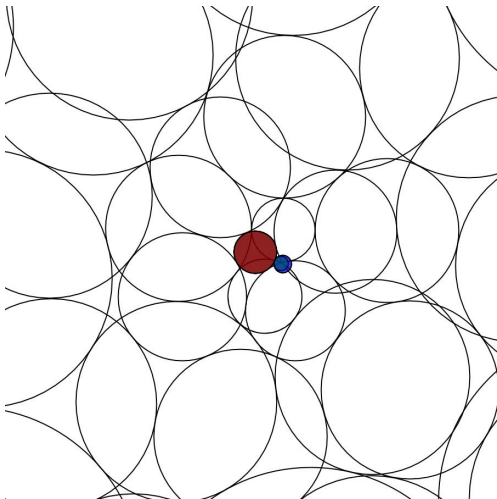
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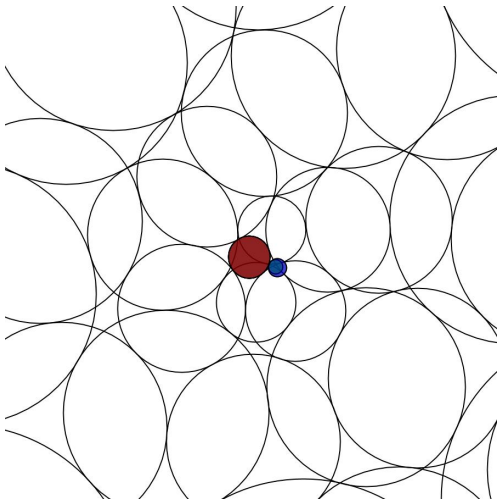
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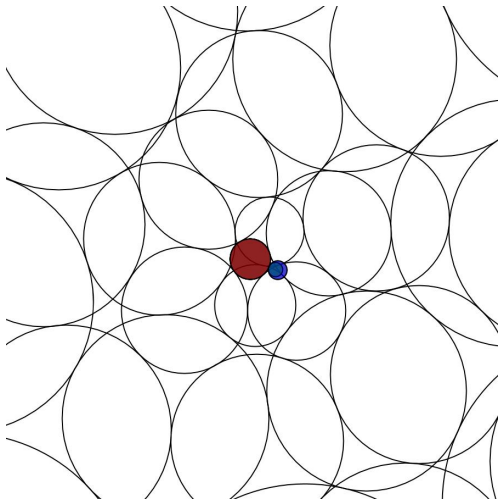
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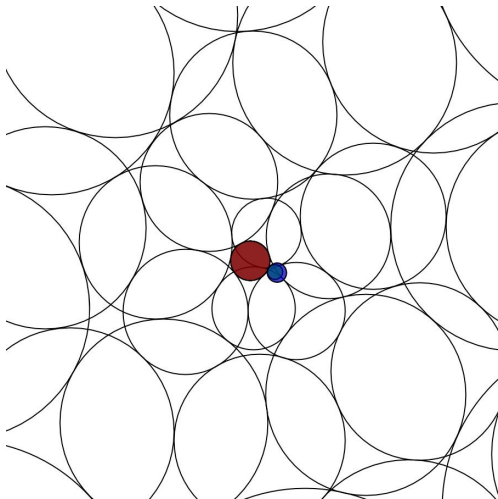
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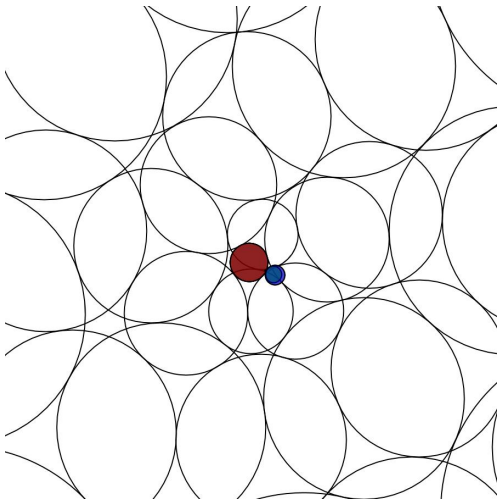
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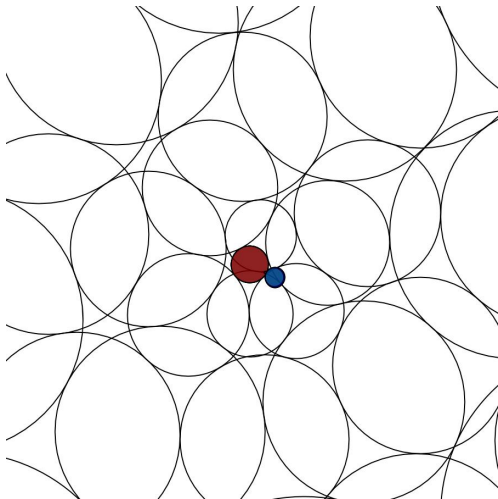
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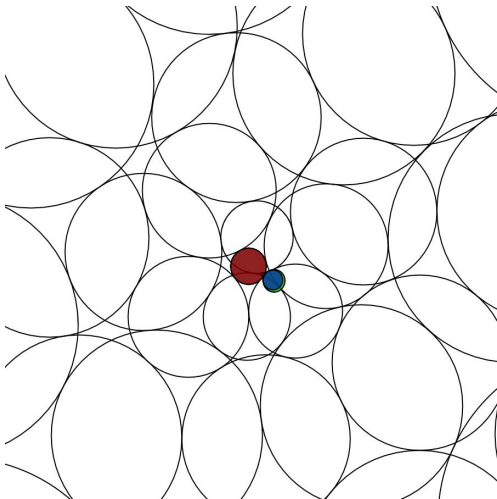
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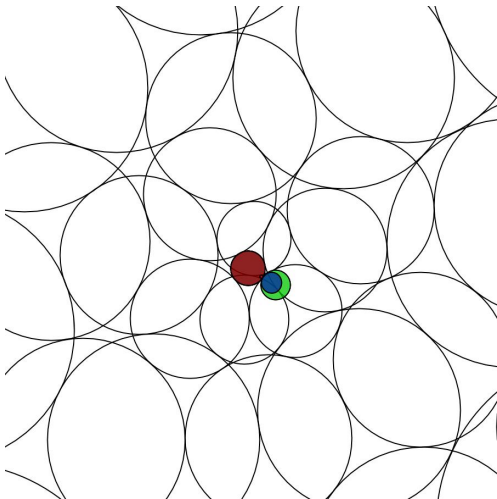


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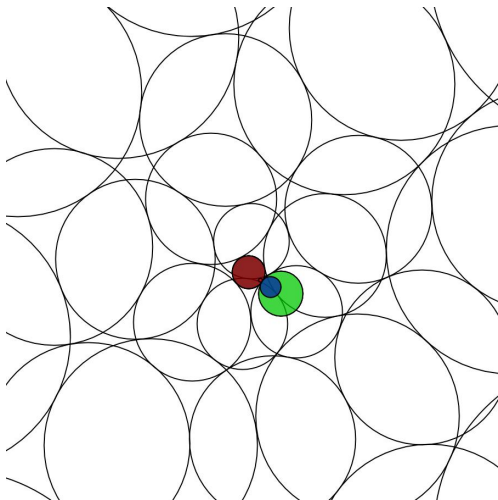
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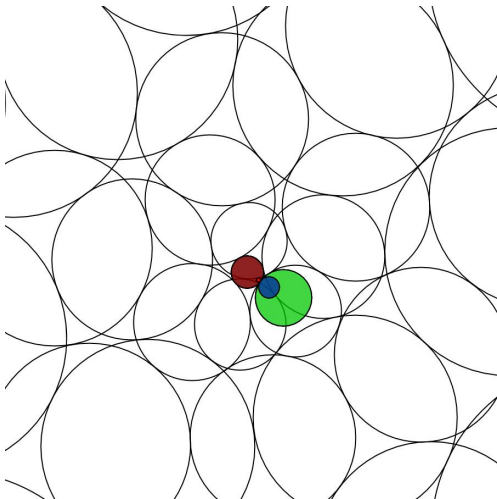
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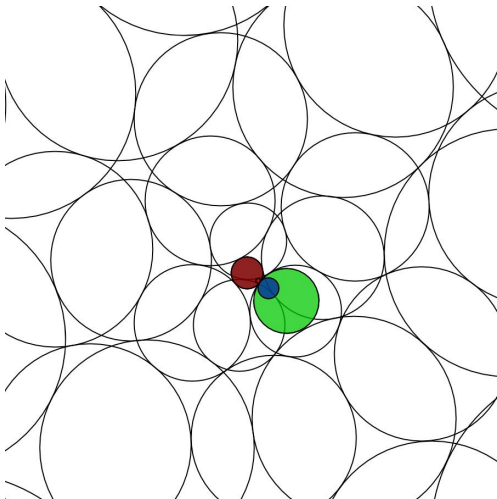
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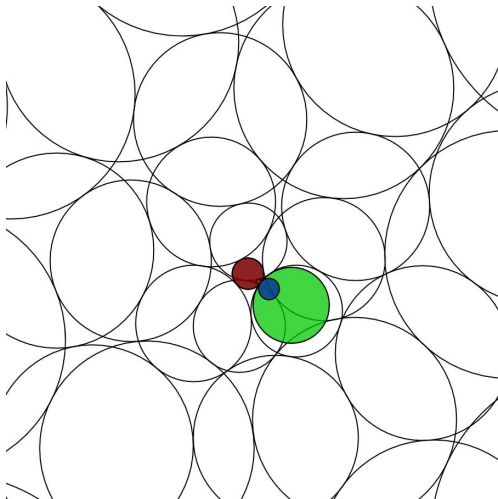
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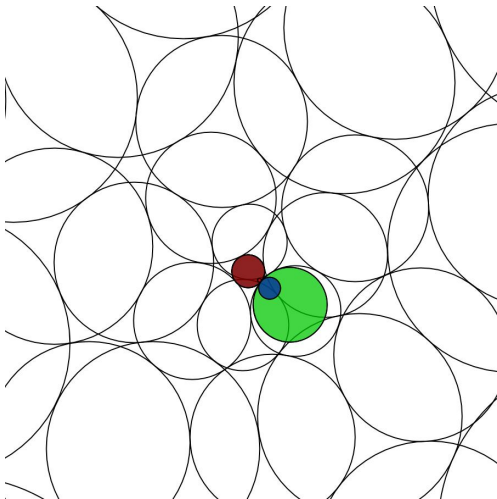
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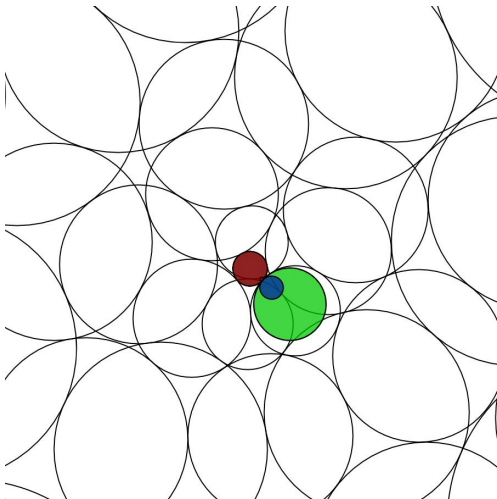
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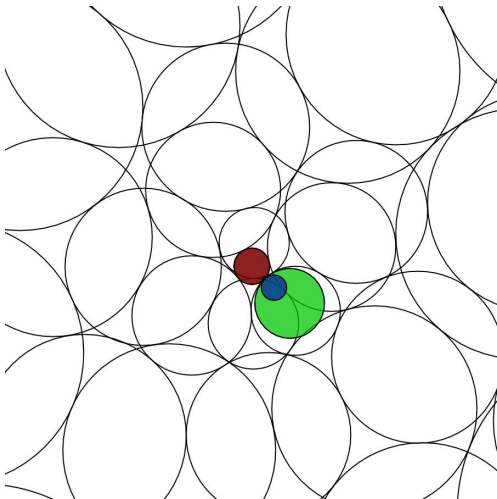
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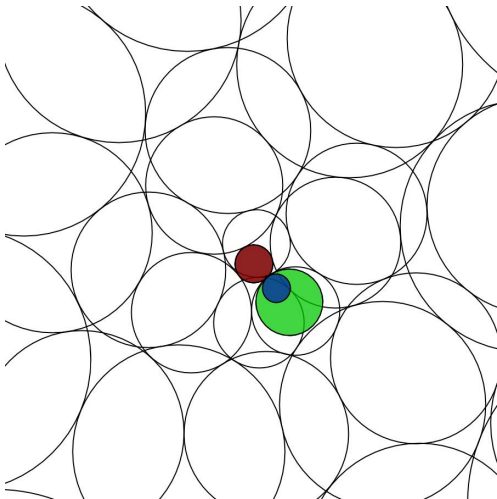


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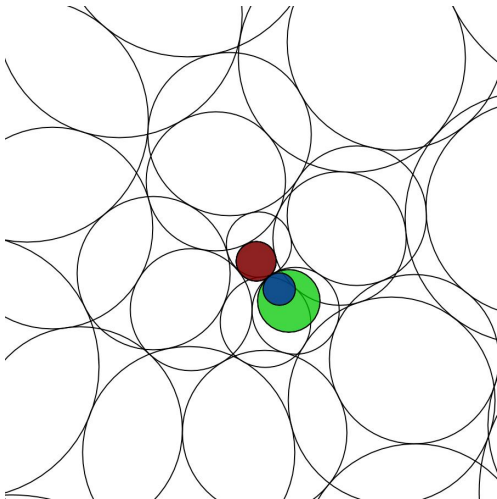
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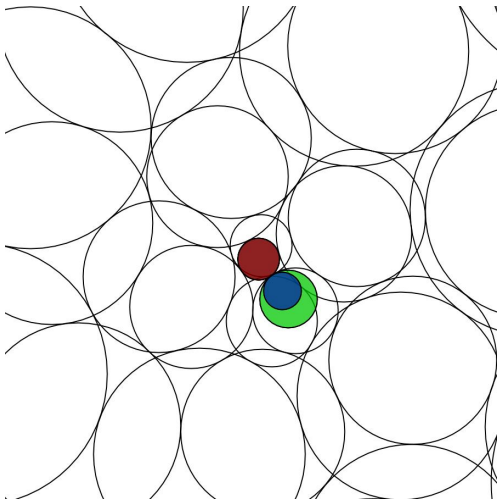
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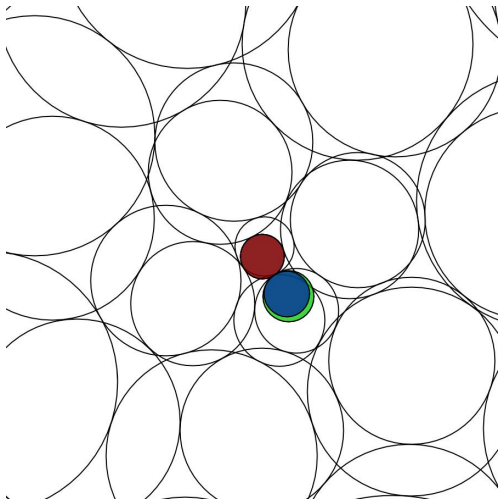
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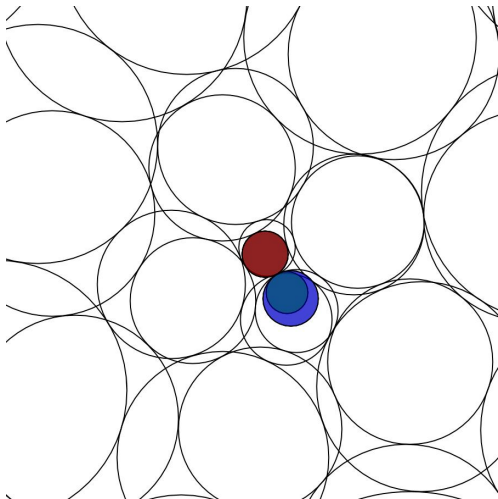
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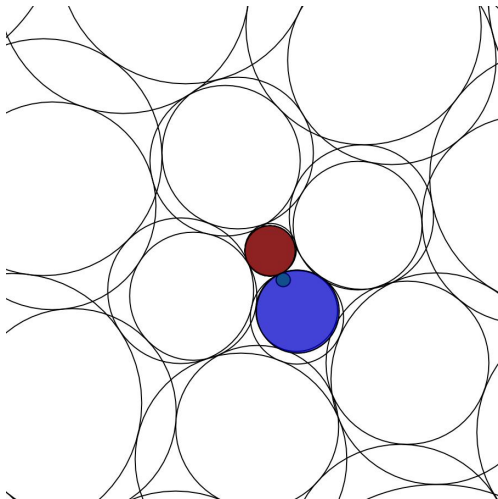
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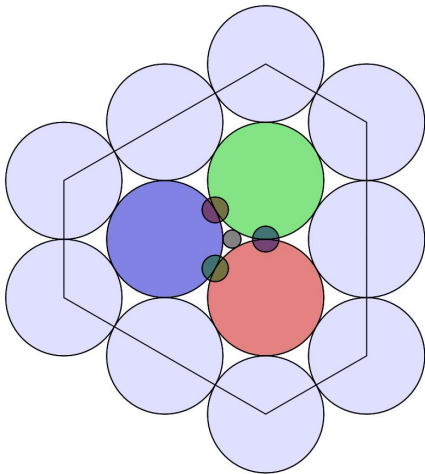
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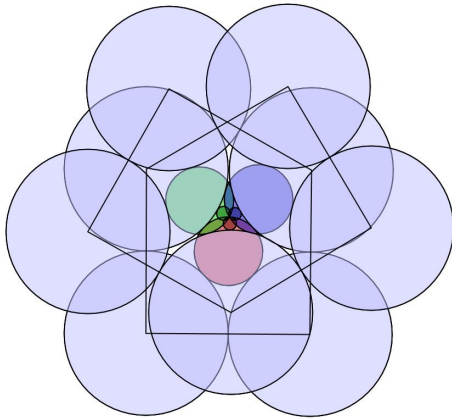
# Singular Branching (interstice of face)



- Auxiliary circles are added, including one at the “branch point”.
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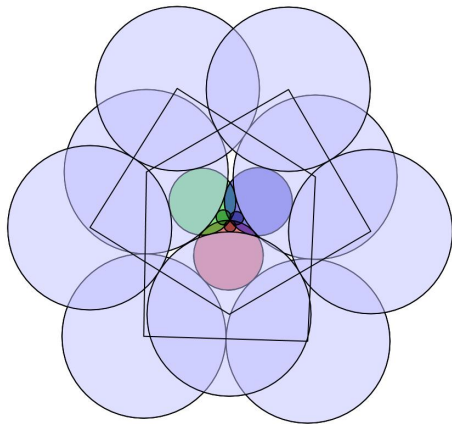


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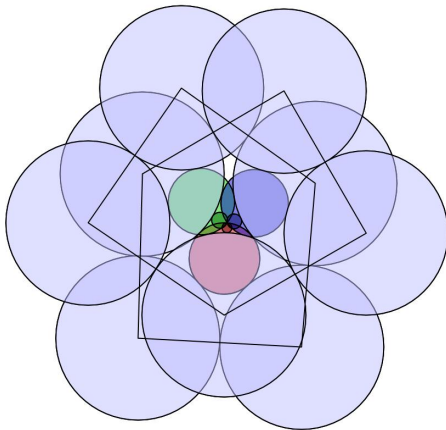
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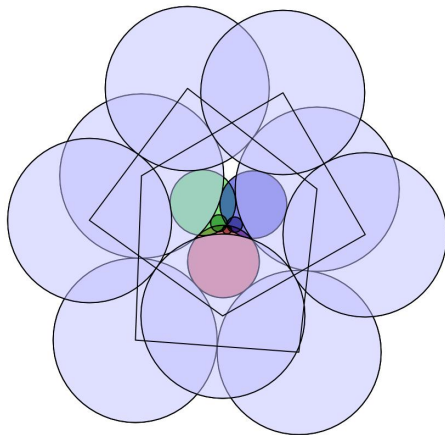
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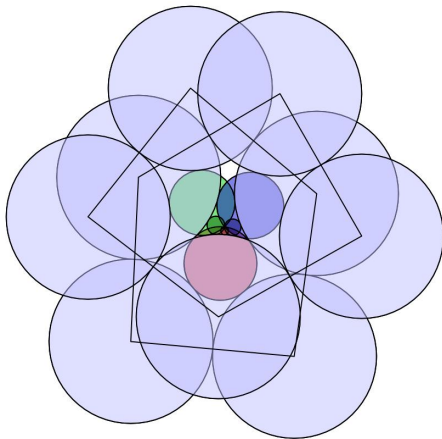
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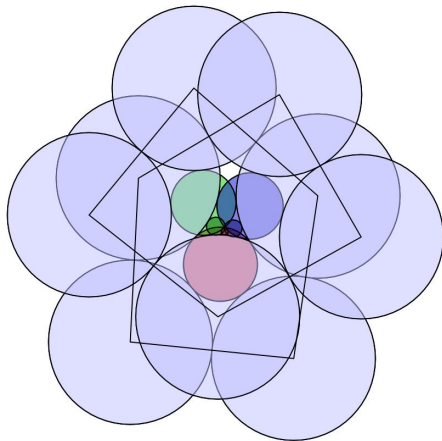
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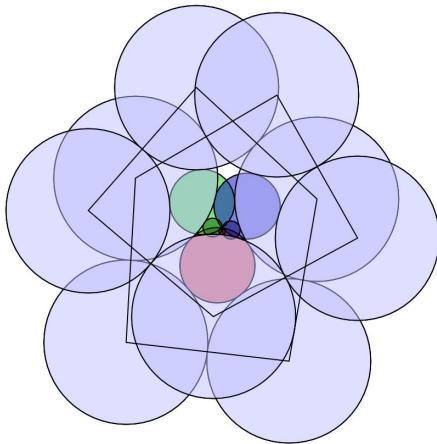
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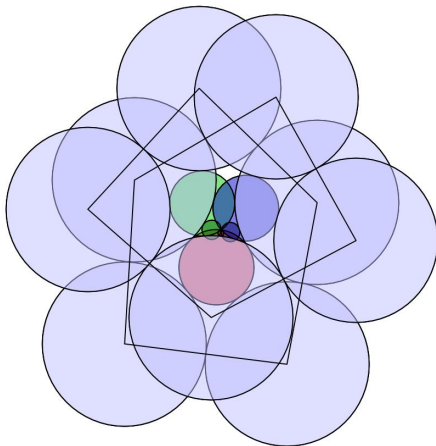
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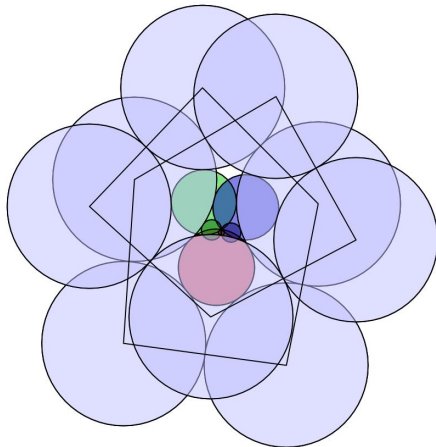
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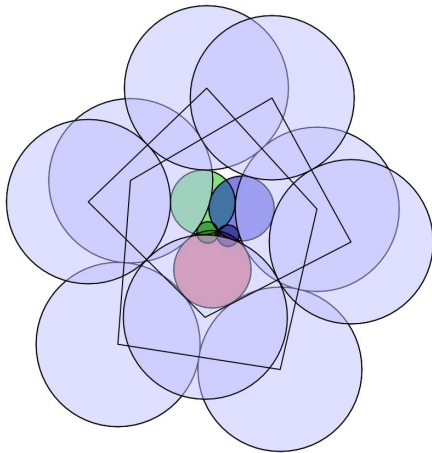


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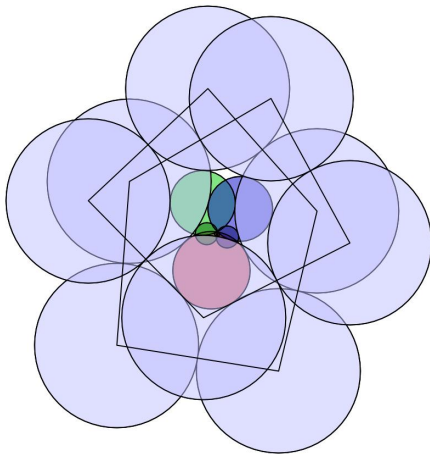
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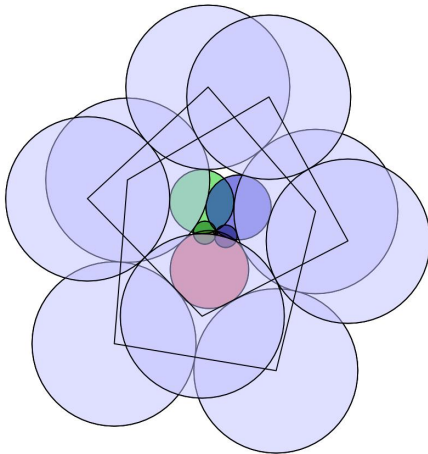
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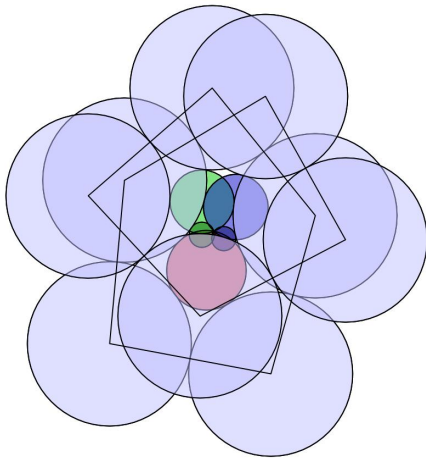
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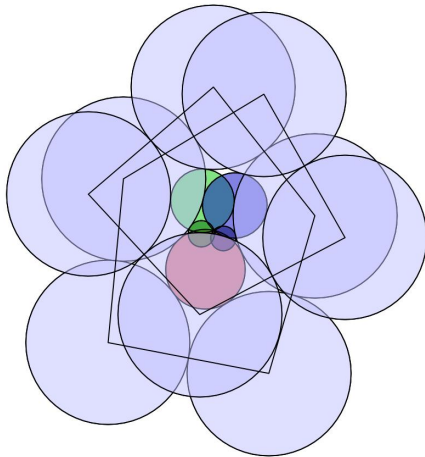
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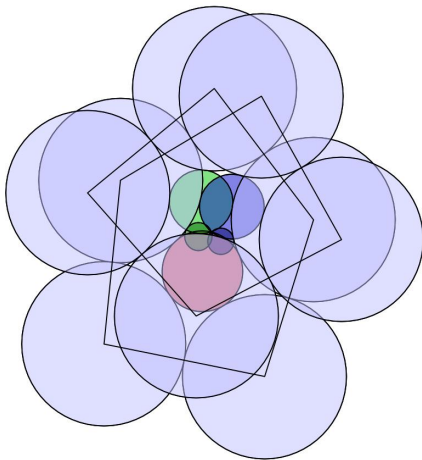
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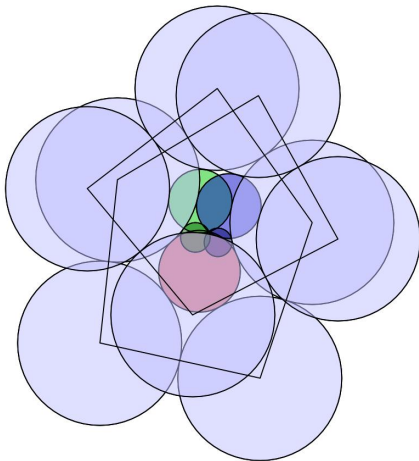
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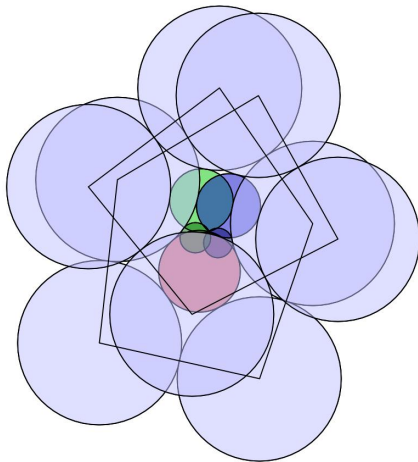
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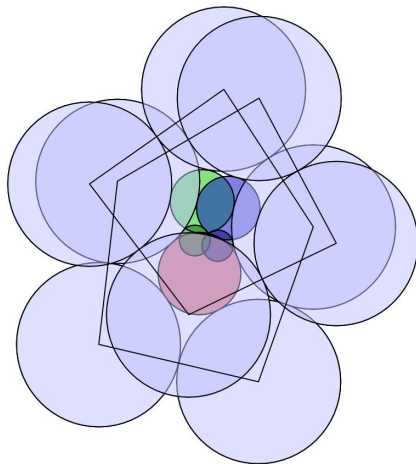


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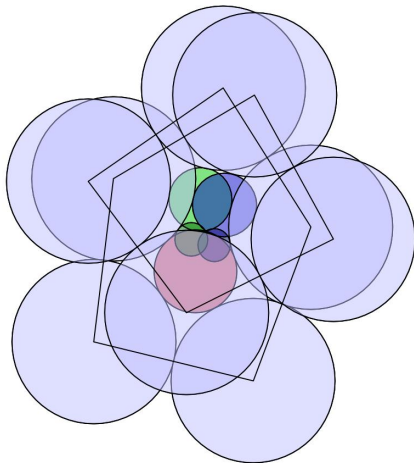
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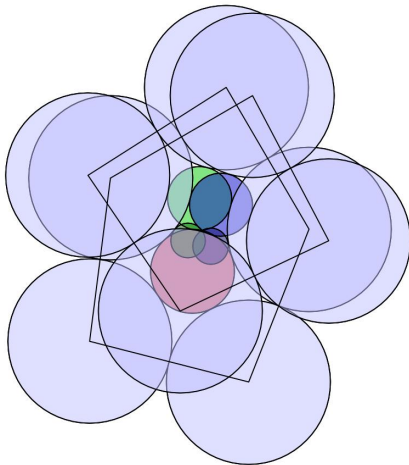
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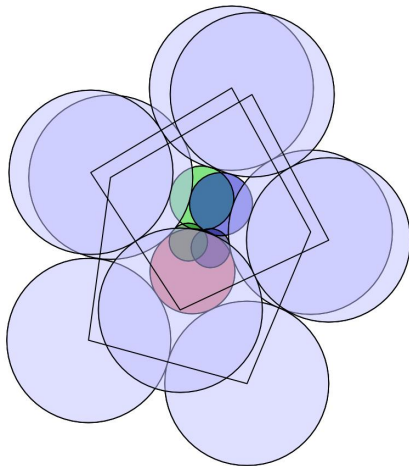
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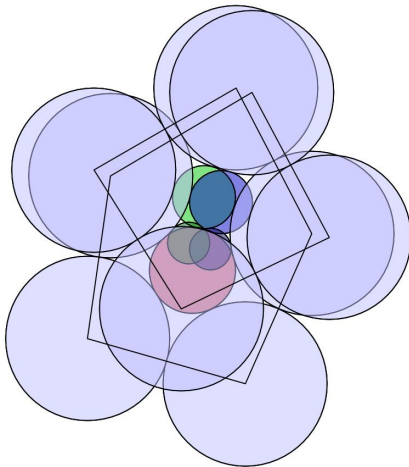
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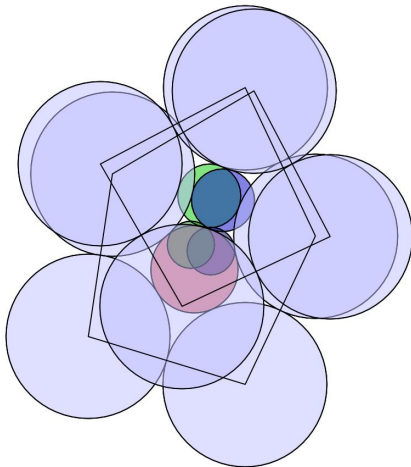
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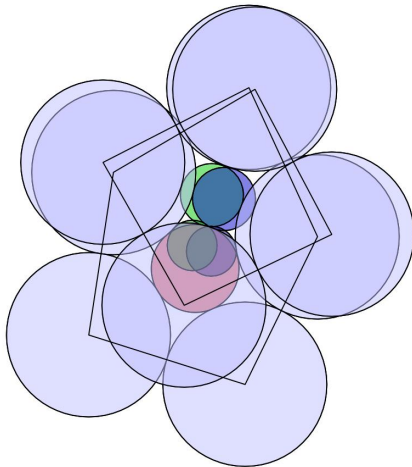
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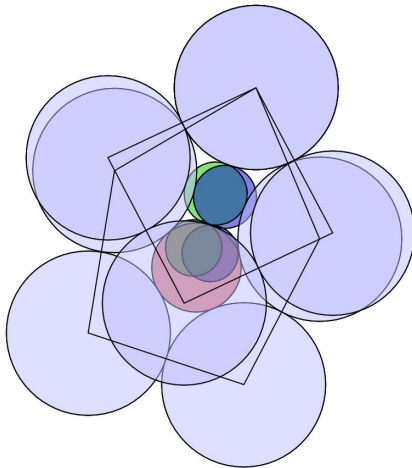
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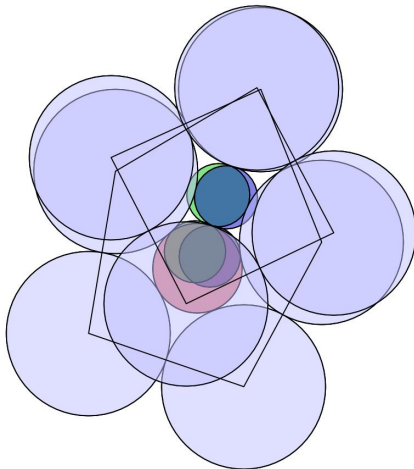


# Singular Branching (interstice of face)



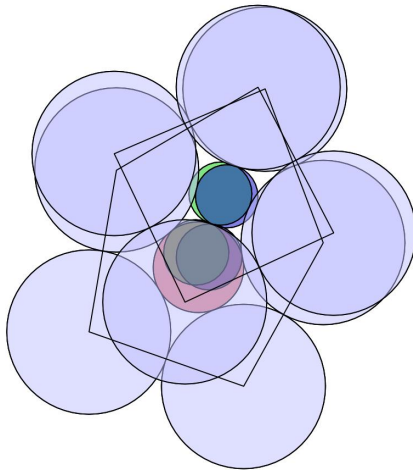
- Auxiliary circles are added, including one at the “branch point”.
- “branch circle” radius is forced to 0.
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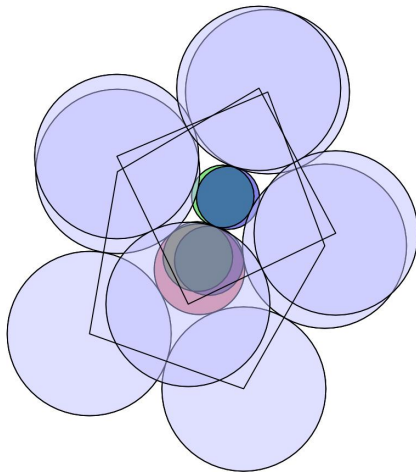
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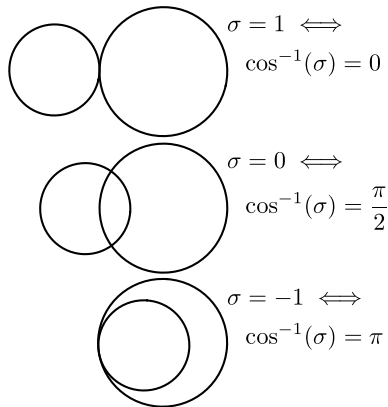
# Triangles with deep overlaps

## Monotonicity Lemma

If the edge assignments of a face are  
in  $[0, \pi/2]$

OR sum to  $\leq \pi$ ,

then for any three radii in  $\mathbb{C}$  or  $\mathbb{D}$ :



# Triangles with deep overlaps

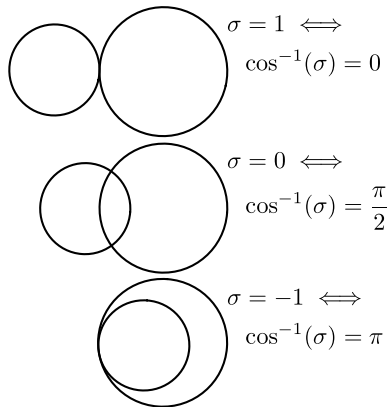
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# Triangles with deep overlaps

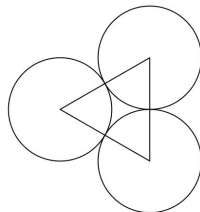
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1. A triangle with these overlaps exists.
2. The triangle is monotone.



# Triangles with deep overlaps

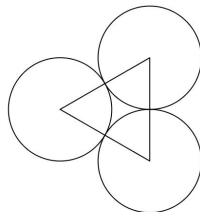
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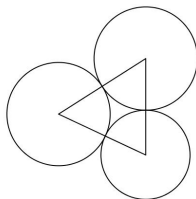
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# Triangles with deep overlaps

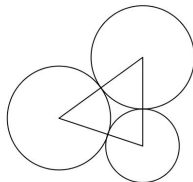
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# Triangles with deep overlaps

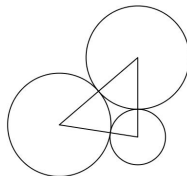
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# Triangles with deep overlaps

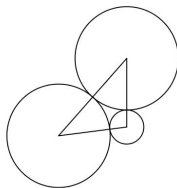
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# Triangles with deep overlaps

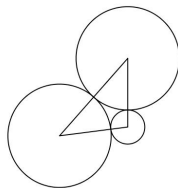
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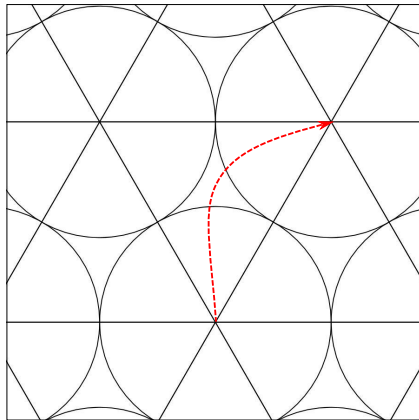
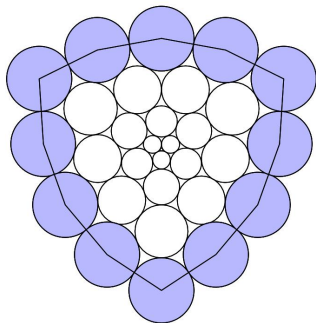


## Theorem

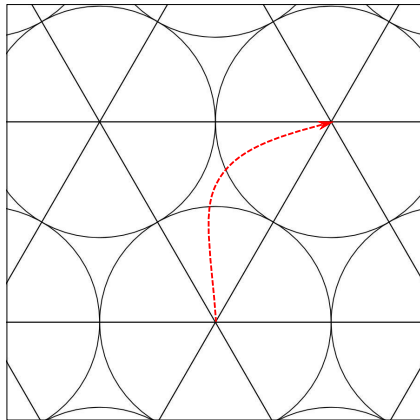
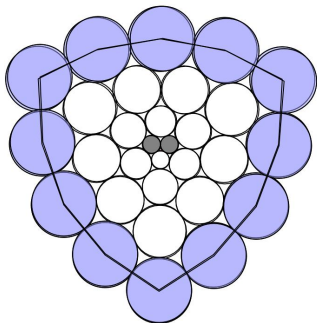
*The radii for a CP with  
singular/chaperone branching exist  
(under some restrictions) and are  
unique.*

# Continuously moving a branch point

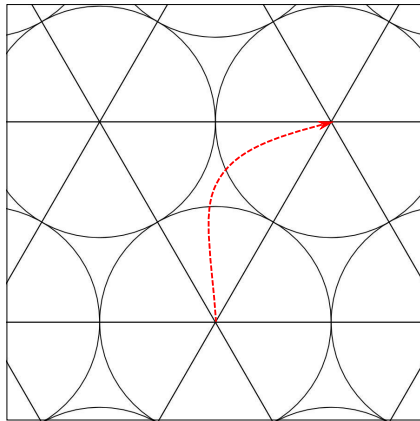
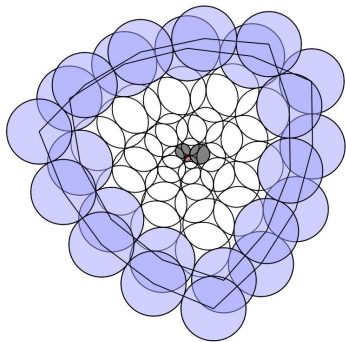
Traditional branching at vertex



# Continuously moving a branch point

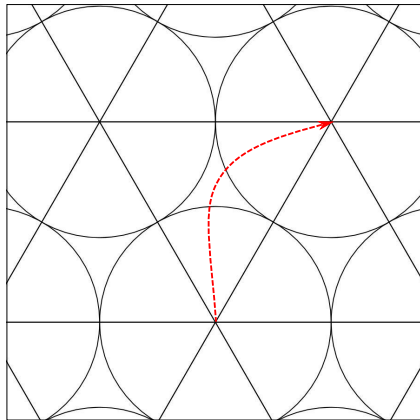
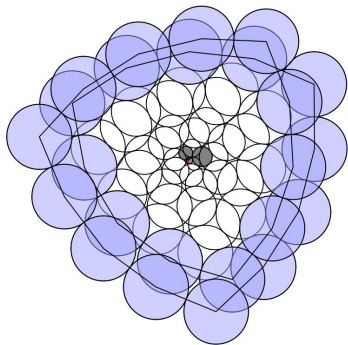


# Continuously moving a branch point

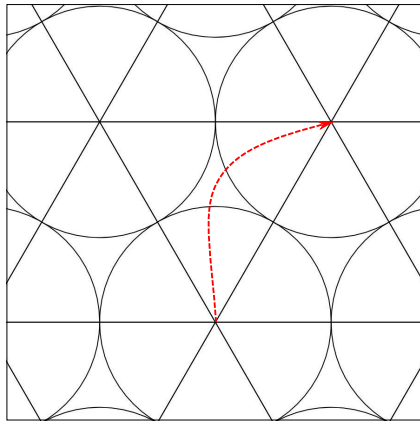
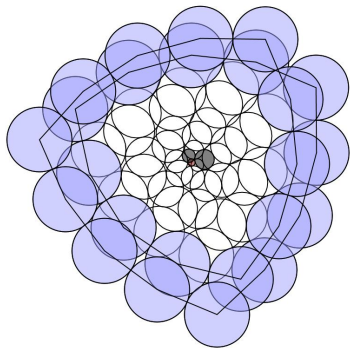




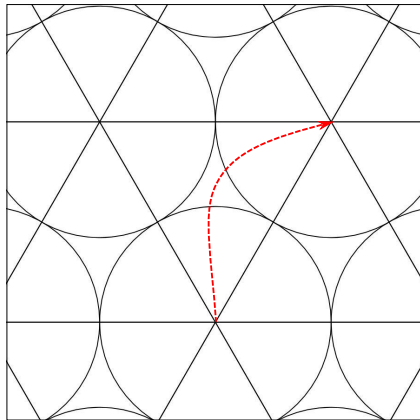
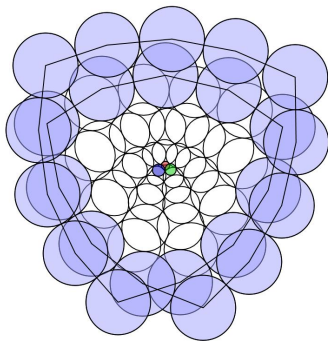
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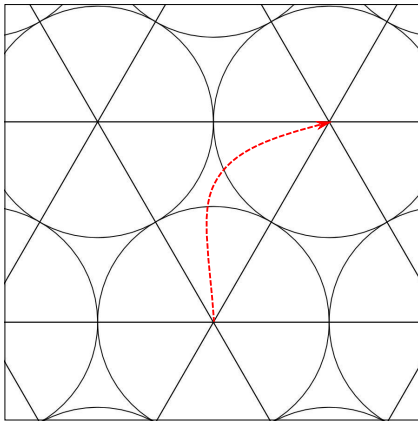
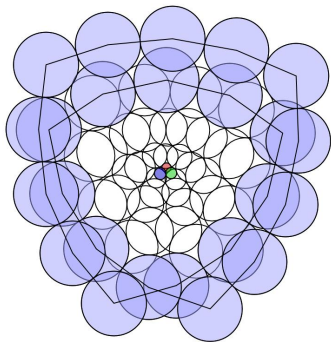
# Continuously moving a branch point



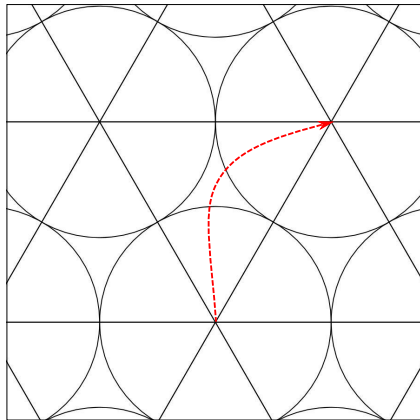
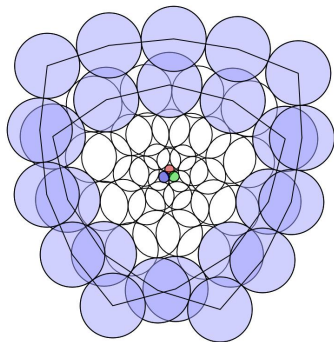
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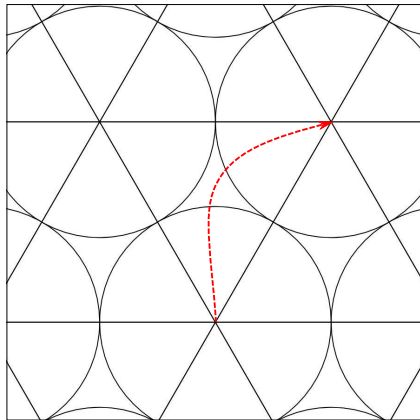
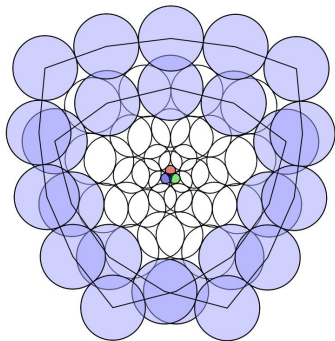
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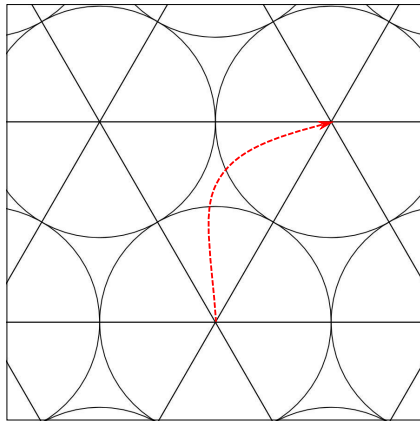
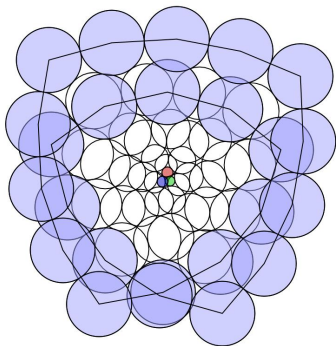
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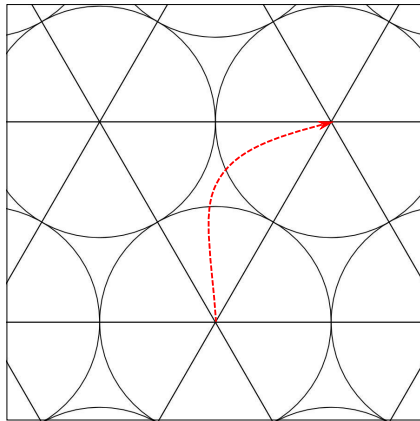
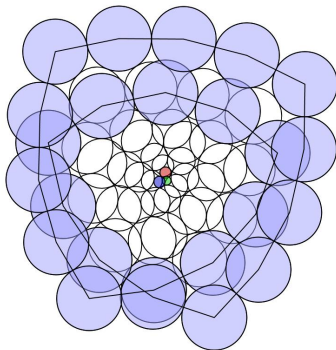
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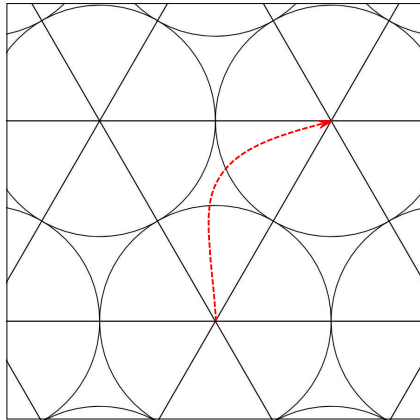
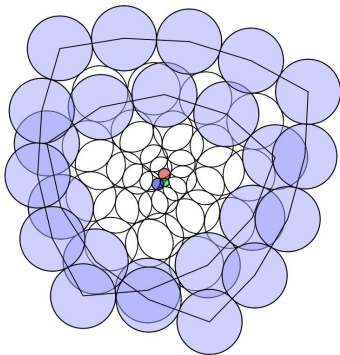


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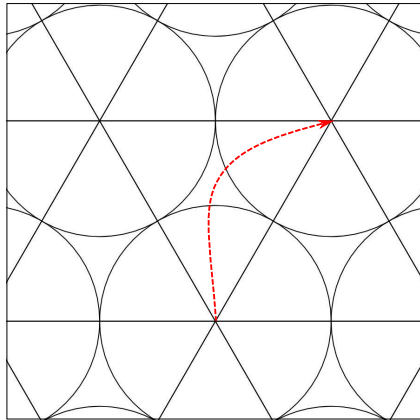
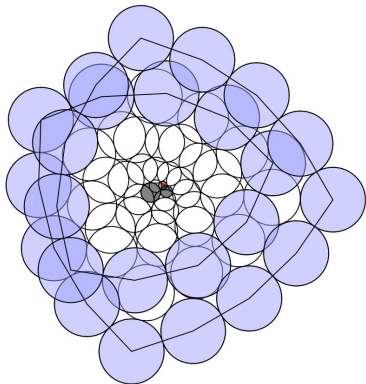




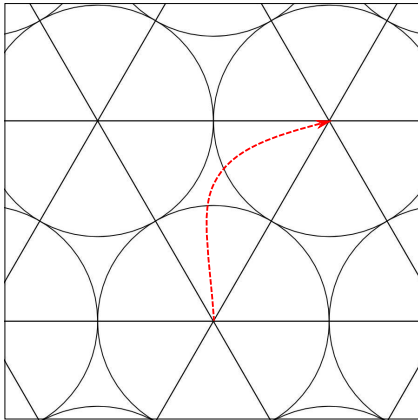
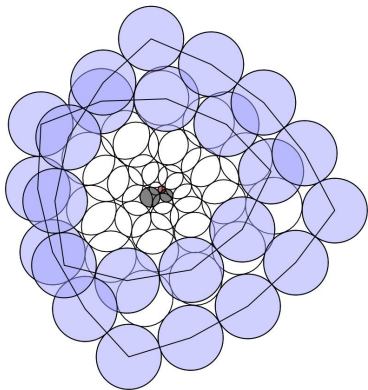
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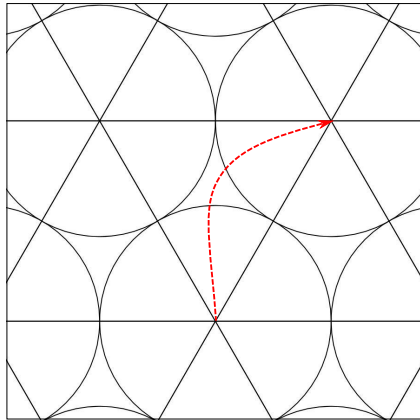
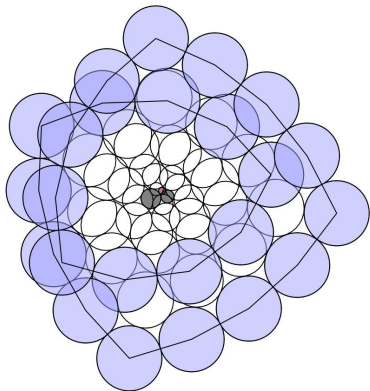
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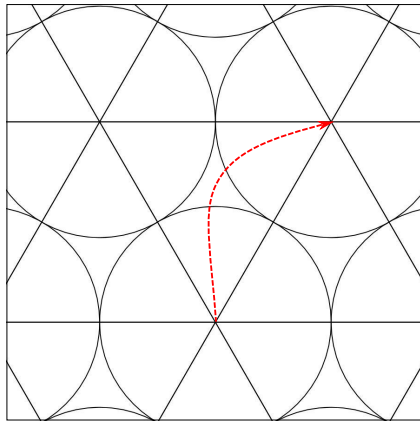
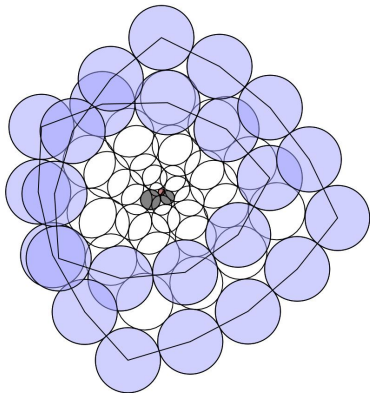
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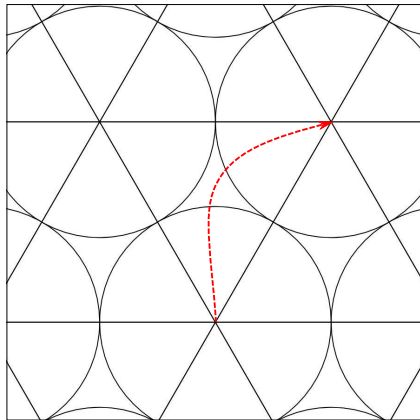
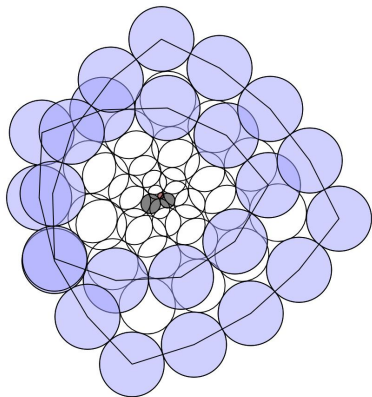
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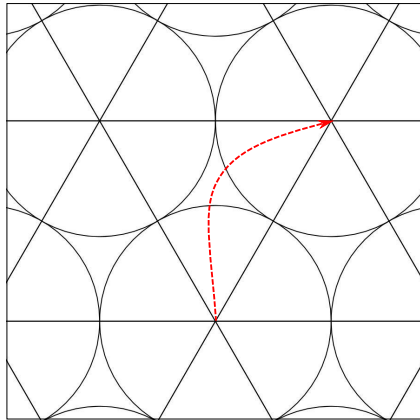
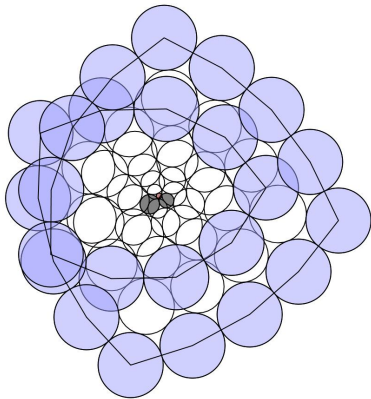
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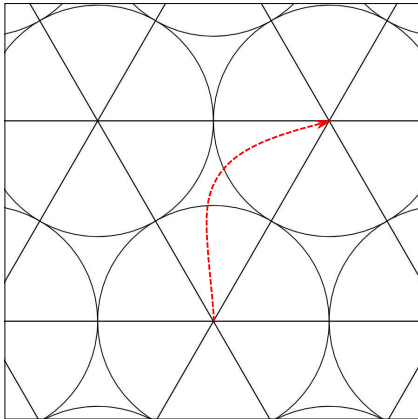
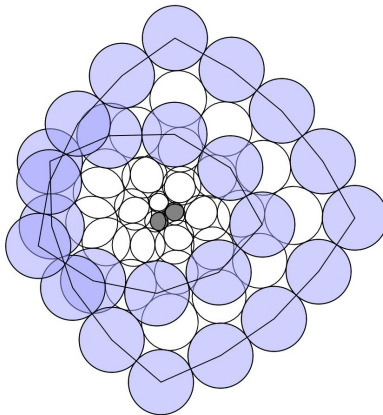
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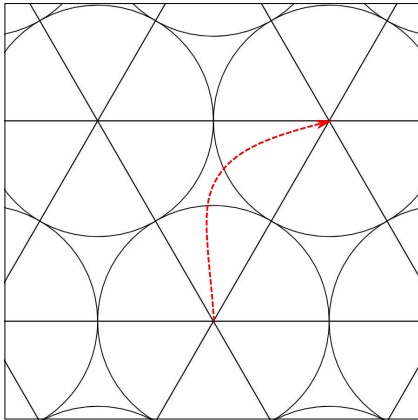
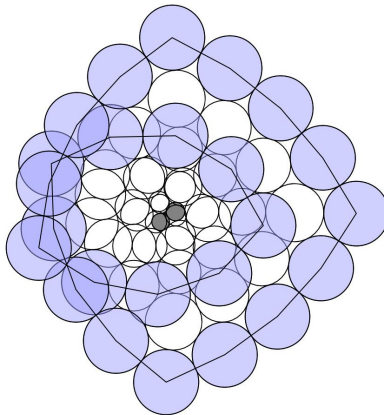


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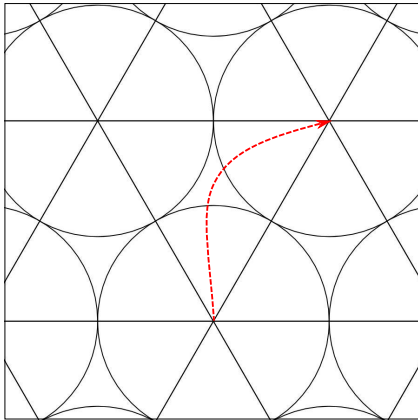
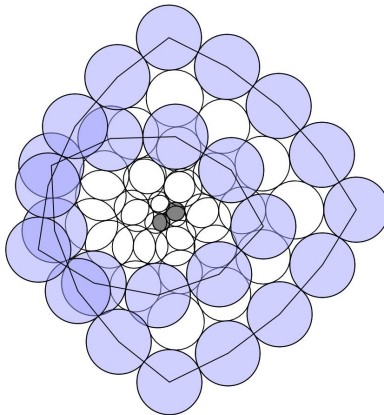




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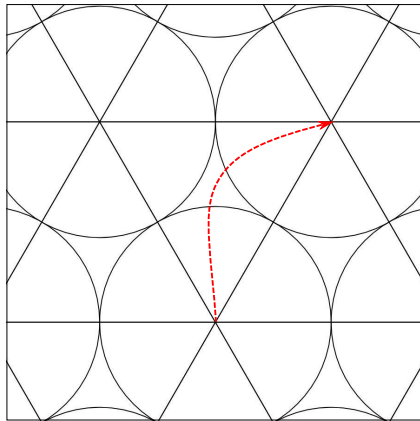
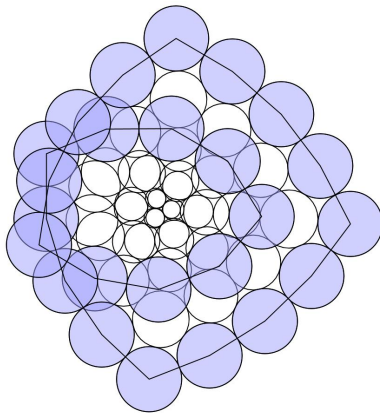


# Continuously moving a branch point



# Continuously moving a branch point

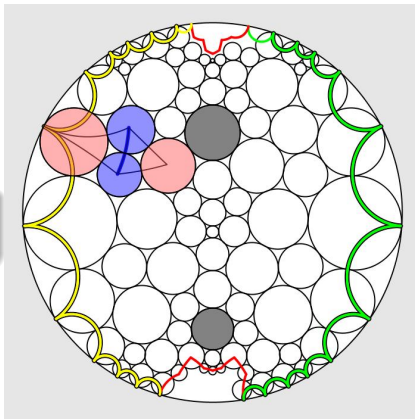
Traditional branching at vertex



# A Discrete Ahlfors Function

## Example

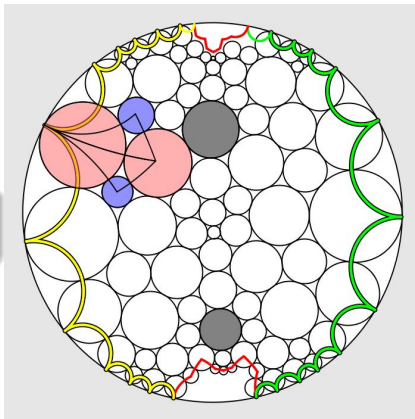
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

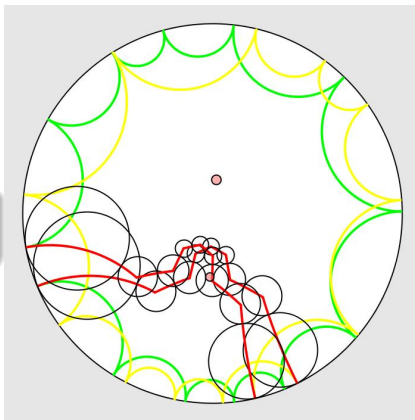
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

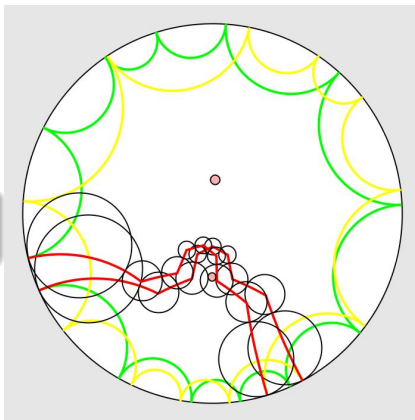
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

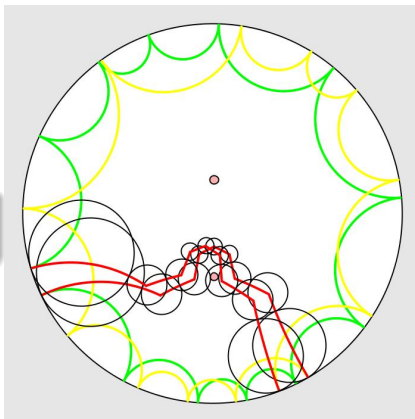
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

A non-symmetric annulus.

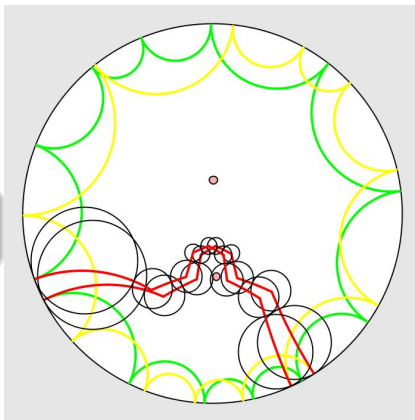




# A Discrete Ahlfors Function

## Example

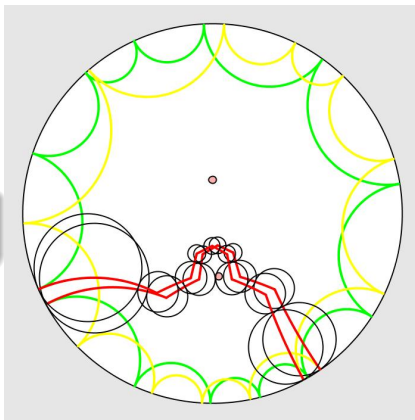
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

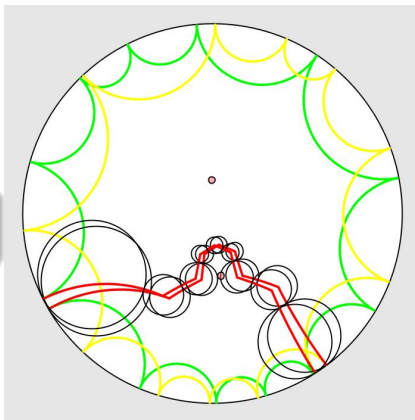
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

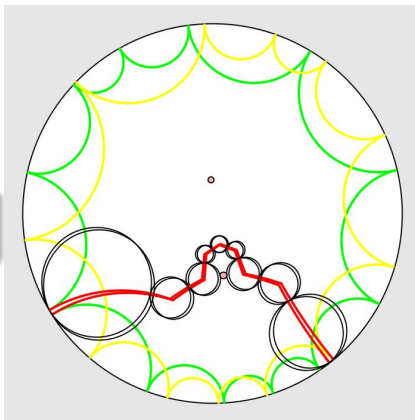
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

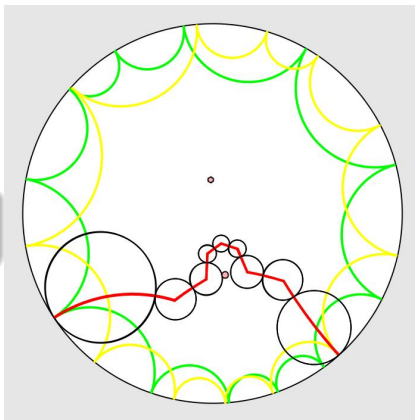
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

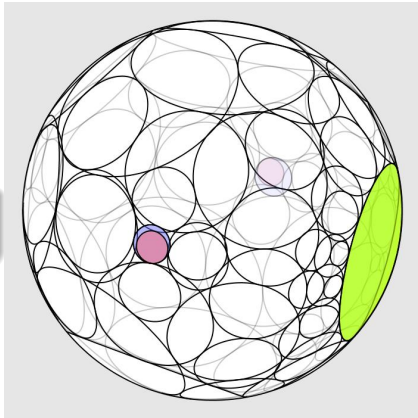
A non-symmetric annulus.



# A Discrete Ahlfors Function

## Example

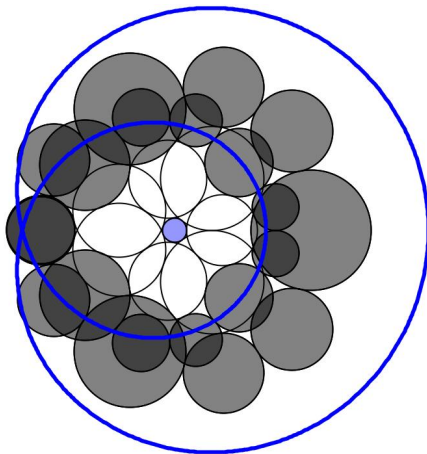
A non-symmetric annulus.



# Another Application

## Example

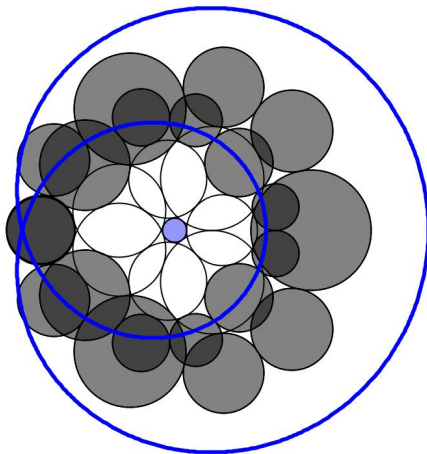
A quadratic polynomial.



# Another Application

## Example

A quadratic polynomial.

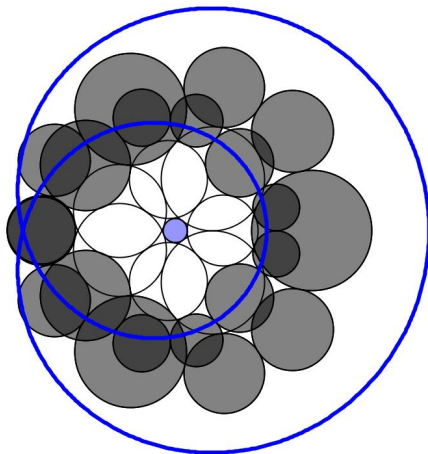




# Another Application

## Example

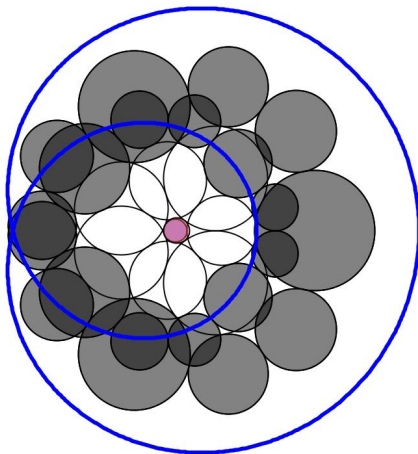
A quadratic polynomial.



# Another Application

## Example

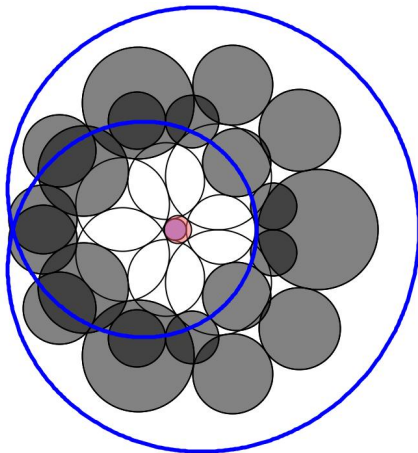
A quadratic polynomial.



# Another Application

## Example

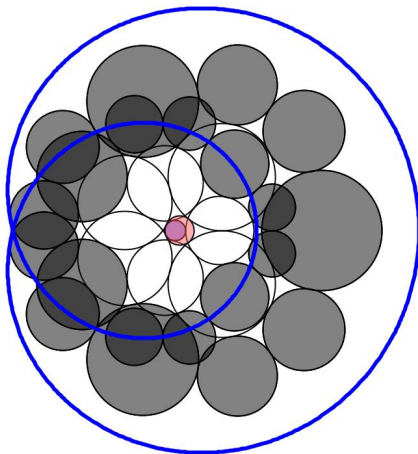
A quadratic polynomial.



# Another Application

## Example

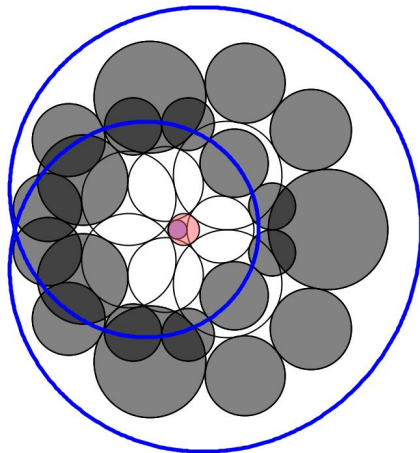
A quadratic polynomial.



# Another Application

## Example

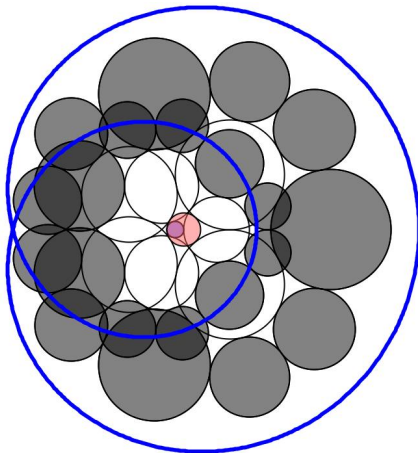
A quadratic polynomial.



# Another Application

## Example

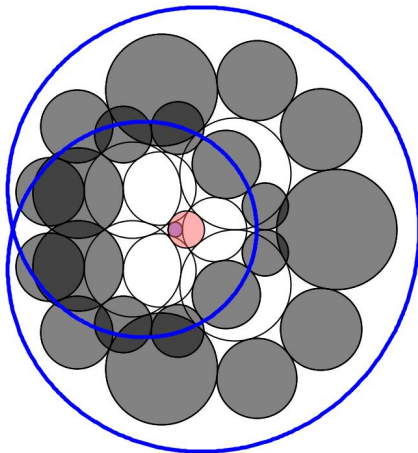
A quadratic polynomial.



# Another Application

## Example

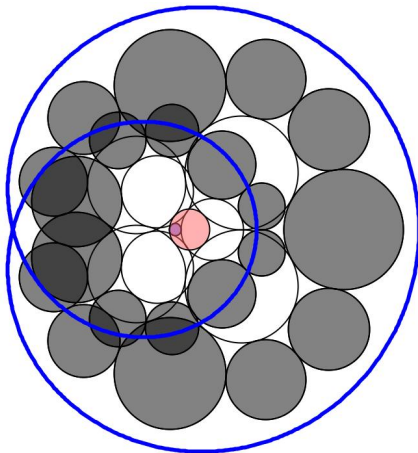
A quadratic polynomial.



# Another Application

## Example

A quadratic polynomial.

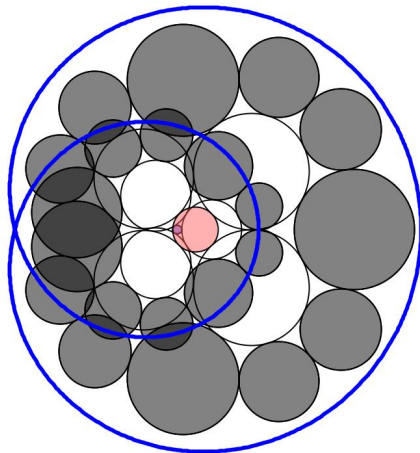




# Another Application

## Example

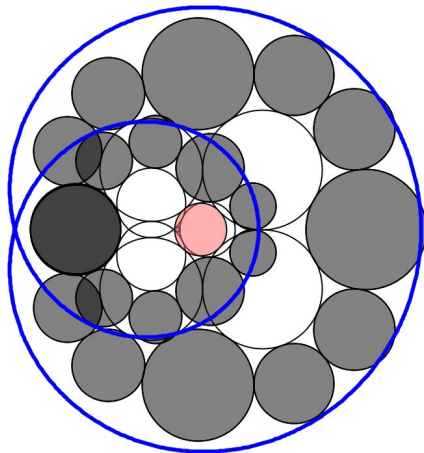
A quadratic polynomial.



# Another Application

## Example

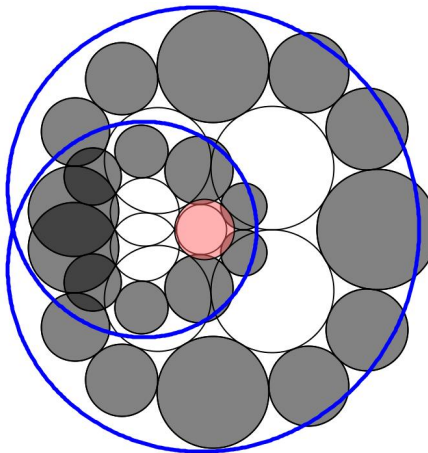
A quadratic polynomial.



# Another Application

## Example

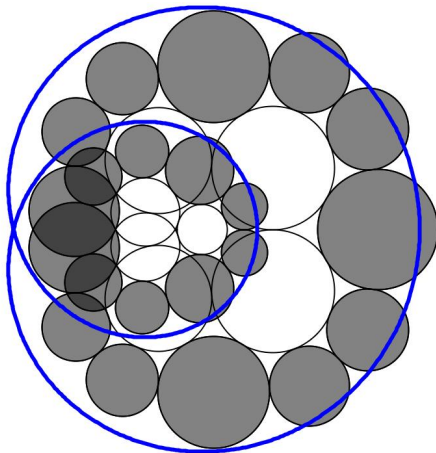
A quadratic polynomial.



# Another Application

## Example

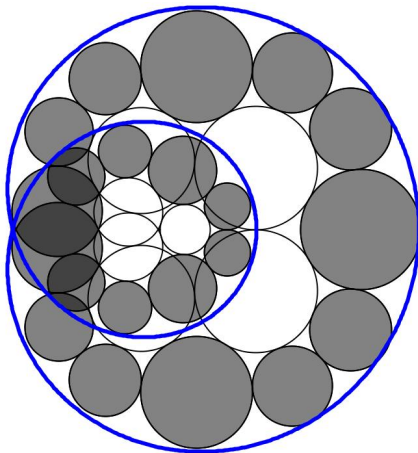
A quadratic polynomial.



# Another Application

## Example

A quadratic polynomial.



All images created with CirclePack

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